National University of Science and Technology

College of Electrical & Mechanical Engineering

(CEME)

Applied Physics

Lab Manual

Department of Basic Sciences & Humanities

Common To

EE, MTS, CE & MECH

College of Electrical & Mechanical Engineering

Department of Basic Sciences & Humanities

Applied Physics

S.NO.	TITLE OF EXPERIMENT
01	Introduction to Lab Equipment: DMM & Power Supply
02	Analysis of Series and Parallel Resistive Circuits.
03	Determination of Resistivity of unknown Material (Wire) using Wheatstone Bridge.
04	Determination of e/m ratio of electron Using Deflection method.
05	Verification of inverse square law by studying variation of photoelectric current with intensity of light.
06	Determination of The Planck's Constant using a Photo Cell.
07	Hook's Law: Determination of Spring constant and effective mass of a spring by static and Dynamic methods.
08	Compound Pendulum: Determination of radius of gyration K and acceleration due to gravity g.
09	Introduction to Function generator and Oscilloscope
10	Determination of RC time constant of RC circuit
11	Investigation of frequency response (VC, XC) of capacitor in RC circuit.
12	Determination of Thermal Coefficient of Linear expansion for different metals.
13	Study of forward and reversed biased I_V characteristics of a Diode.
14	Hall Effect: Study the Hall voltage relationship with magnetic field and current.

This handout discusses aspects of the experimental lab work that you will perform in Applied Physics Lab. It' begins with a discussion of good lab practice, and ends with a discussion of lab safety. You should pay special Attention to the Electrical Safety Notice attached to the end of this handout.

Lab Rules:

1. You have to be in time. Students coming in late (more than 15 minutes) must consider the fact that they may not get the permission to admit the Laboratory.
2. Leave your belongings (bags, Jackets etc) on table (placed at the main entrance of laboratory).
3. **Every student will be allocated a group and a workstation which he/she is not supposed to change throughout the semester.**
4. **Use of mobile phones and laptops (unless instructed) during laboratory session is strictly prohibited.**
5. **Students are not allowed to move or replace the equipment designated to each workstation**
6. Eating, drinking, smoking, using the mobile phones is not allowed in the Laboratory.
7. Both your safety and the safety of others depends on your behavior and attitude.
8. Listen to **ALL** instructions given by the supervisors and follow them carefully.
9. All students are obligated to have the valid student identity cards during the classes.
10. Students may not leave their workspace without the assistant's permission. Especially they may not go around the Laboratory, disturbing other groups.
11. Notify your supervisor of any accident regardless of its importance. If you are injured inform the assistant to receive the first aid immediately.
12. Do not apply voltage to the electric circuits until controlled by the supervisor.
13. Do not modify the measurement setups.
14. The Student are instructed to show your results and get their work signed by instructor before leaving the lab.
15. Switch all power supplies off and make an order after your measurement.

Pre-LabWork

Pre-lab work is designed to motivate and define your experiments and thus prepare you for your in-lab work. Pre-lab work should be treated like a small homework, and should be written up in your lab notebook. The questions asked in the pre-lab assignments are indicative of. The type of questions you should ask when examining a scientific or engineering hypothesis. Your analysis of these questions should motivate and guide the experiments you will perform, and suggest how the experimental results will confirm or refute the hypothesis under examination.

The pre-lab assignments will generally analyze and predict the performance of an Physics

circuit; and result in predictions for the data you will record during your in-lab work. As you work through type assignments, you should think about how you could experimentally determine whether your analysis is correct. If you do not see how the in-lab work seeks to verify your prelab work then I} either effort is of value even if both are correctly performed. Finally, as you work on the pre-lab assignments, you would take the time to draw the circuits you will build during your in-lab work, and to prepare any graphs and tables necessary to organize data recording.

In-LabWork

The notes you take during your in-lab work are simply a record of what you did and what you observed. A minimal record includes at least the following.

1. A labeled circuit diagram in your lab notebook adjacent to the recorded data. The diagram should include voltage and current source amplitudes and frequencies: resistor, capacitor, and inductor values; semiconductor and integrated circuit names; oscilloscope and multi-meter. How connections you think are important to circuit operation such as shielding, component temperature and so on. You should also note or sketch your predictions before you take ameasurement.
2. A record of your specific procedures and measurements. This should include key oscilloscope and multi-meter settings, source adjustment made while recording data, small component changes and so on. You should record raw data in your notebook and interpret it later, even if later means only a few minutes later before you move on to the next experiment: For example, if you use voltage drop across a resistor to measure a current you should record the measured voltages in your lab notebook and convert these measurements to current later. If any data is unexpectedly large or small, or noisy or free, for example, or some adjustment is particularly large or sensitive, you should note this in your lab notebook as part of the data.
3. Finally, included in your notebook should be all comments necessary to recreate your experimental procedure. Waveforms sketched directly from the oscilloscope warrant a few additional guidelines. Never make a rough sketch that you will copy into your lab notebook later: each time data is copied from one medium to another, new errors and interpretations creep silently into the data. Rather, neatly sketch the waveform directly into your lab notebook as accurately as possible. Such sketches should be at least as large as the oscilloscope screen, and are most accurate if you let the ruled squares of your notebook graph paper correspond to the ruled gratitude on the oscilloscope screen.
4. Commentary about inaccuracies in your data. Record and plot precisely what you see as accurately as possible; do not record what you expect to see. Related to the accurate recording ofdata is the issue of precision. You must indicate in your data how precise your readings are. Such an indication can be as simple as an estimate of the noise in the

measurement. When sketching oscilloscope data be sure to indicate the trace-width and any other relevant features of the waveform.

Post-LabWork

Post-lab work concerns your interpretation of your results in terms of the analysis and prediction of your pre-lab work. It is the time to discuss not only what the data tells you about the circuit operation but also whetl,1er the data answers your questions from the pre-lab, and supports or refutes any hypothesis you have made. Important questions to ask and answer in the notebook include at least the following.

1. Did this data match the predictions? Why or why not?
2. What does the data say about the theories used to drive the predictions?
3. Under what conditions do the theories apply and why?
4. Where those conditions met during the in-lab work?
5. What factors could be generating what appear to be bad data?
6. How might the pre-lab analysis be modified to make more accurate predictions in the future?
7. How could the experiments be modified to obtain data that are more accurate?
8. The post-lab work is also the time to combine data from different experiments in order to reach more general conclusions.

Lab Report Format(Only handwritten reports will be accepted)

I. Cover Page:

a. Degree, Discipline and syndicate (0.25)

b. Group No(0.25)

c. Students Name with Reg. No.

d. Experiment No. and Title

e. Due Date

f. Date of Submission

II. **Objective:**

State the objective clearly for the lab task.

III. **Theoretical Background**: Mention theoretical knowledge required for the experiment along with mathematical models and laws.

IV. **Materials and Equipment**:

V. **Procedure**: Briefly write in points the procedure you followed during lab.

VI. **Theoretical Data**: Include any data you can predict using mathematical models and laws.

VII. **Laboratory Data:** This section is for your collected data, alongwith any plots, tables, or illustrations.

VIII. **Comments and Conclusions:** Discuss what went as expected and what did not. Compare your theoretical and laboratory results. Are they same? Why or why not.

IX. **Troubleshooting:** Write the problem faced during experiment and how you detect and resolve that.

X. **Questions:** Make sure that if there are any questions in the lab manual are answered completely

Be creative. Your lab report should not be copied. After due date reports will be not accepted.

Oscilloscope Readings

Make your measurement at the middle of the trace. A widened trace is most often caused by added noise. In this case, the reasonable assumption that the average of the noise is zero leads to the conclusion that the true value of the signal lies at the middle of the trace. This is particularly important when measuring the peak to peak voltage of a periodic signal. Even in the presence of a little noise, measuring from the extremes of the trace can easily add 3% to 5% to the measurement, thereby reducing its precision to less than two digits.

The oscilloscope is an extremely versatile measuring instrument and you should attempt to take advantage of its capabilities. Adjust the voltage scale so that the waveform fills the scope screen and the effects of trace width are minimized: In general, do not hesitate to move the trace around on the screen or expand interesting parts of the waveform in order to get a more accurate view of its detailed behavior.

Bread boarding Practice

Following the bread boarding practice outlined below will generally help guarantee that your circuit will work as well as possible.

(1) Lay the circuit out as neatly as possible. Having the component locations correspond to the circuit diagram will aid in Iodating test points. Such, a layout is also easiest to check for wiring errors.
(2) Make all wires and leads as short as possible. Long wires and leads increase stray capacitance and inductance which can be significant in high-frequency circuits.

(3) Take care when inserting and removing integrated circuits, to and from the breadboard so as to avoid bending the pins. To remove a DIP, insert a screwdriver or other thin object under the package and pry up gently. It is best to pry a little on alternate sides so the pins are not bent in

any direction. Never pull out an integrated circuit or other circuit element with the power connected.
(4) If your circuit exhibits significant noise, significant distortion, or poor high-frequency or performance, try by-passing the power supplies by placing 0.1µf capacitors across the positive supplies. Also, place the capacitors across the power supply terminals of all integrated circuits.

SAFETY

Safety is a very important component of good lab practice. According, please read, and follows the attached Electrical Safety Notice .As indicated, it is necessary to sign the last page of the notice and return it to the PHYSICS LAB Instrument Desk in order for you to pick up your lab kit.

ELECTRICALSAFETY

For Staff and Students in the Lab

NEVER WORK ALONE

If you are working with energized circuit~ or equipment over 50 volts peak, make sure that at least one other person can see you and hear you, In case of emergency notify the Lab Assistant on duty.

VOLTAGE RULES

All Physics lab Instrument voltages are below 50 volts peak. If you intend to work on a project using power sources over 50 volts peak, you must secure permission and receive specific training from your Instructor, TA, or Lab Technical Personnel before any work on the project begins.
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PREVENT ACCIDENTS: FOLLOW THIS ADVICE

- Never hurry. Work deliberately and carefully.

- Connect to the power source LAST.If you are working with a lab instrument that has internal power supplies, turn the main: **power switch OFF** before you begin work on the circuits. Wait a few second~ for power supply capacitors to discharge. These steps will also help prevent damage to circuits.

- If-you are working with a circuit that will be connected to an external power supply turn the power switch of the external supply OFF before you begin work on the circuit.

• Check circuit power supply voltages for proper value and for type (DC, AC, frequency) before energizing the Circuit.

• Do not run wires over moving or rotating equipment or on the floor or string them across Walkways from bench-to-bench.

• Remove conductive watch bands or chains, finger rings wrist watches, etc. and do not use metallic pencils, metal or metal edge rulers, etc~ when working with exposed circuits.

• When breaking an inductive circuit open the switch with your left hand and turn your face away to avoid danger from any arc which may occur across the switch terminals. When using large electrolytic capacitors be sure to wan long enough (approximately five times constants) for the capacitors to discharge before working on the circuit. All conducting surfaces intended to be at ground potential should be connected together.

ADDITIONAL CAUTIONS

The Physics Lab is equipped with circuit breakers. Check for leakage path to ground when breakers trip repeatedly and the problem is not due to an overload.

• Any equipment used in the laboratories must be equipped with a standard three-prong AC plug or a two-pronged polarized plug.

•All exposed non-current-carrying metal parts of fixed and portable equipment that may accidentally become energized should be grounded.

• All electrical equipment or apparatus that may require frequent maintenance must be capable of being completely disconnected from the power source.

• Do not bring into the lab or use in the lab equipment that does not conform to these rules. Without specific permission *from* your instructor, TA, or Lab Technical Personnel.

QUESTIONS ABOUT WORK/LAB SAFTELY

Any questions about work or lab safety should be brought to the attention. of your immediate supervisor or instructor. If problems arise- that cannot be solved at this level, you should contact the Lab Engineer / Lab OIC.

BASIC ELECTRICAL SAFETY PRACTICES

The Physics Lab requires everyone who uses electrical equipment to understand these safety Precautions. The following safe work practices an prevent electrical shock. Contact your

Supervisor for additional safety training if your job involves repairing, installing or working on energized parts.

A. Safe Work Practices

1. Turn off and unplug equipment (instead of relying on interlocks that can fail) before removing the protective cover to clear a jam, replace a part, adjust or troubleshoot. Ask a qualified person to do the work if it involves opening equipment and creating an exposure to energized parts operating at 50 volts or more.

2. Don‘t use an electrical outlet or switch if the protective cover is ajar cracked or missing Inform Lab Assistant and Lab Engineer to get it fixed.

3. Only use DRY hands and tools and stand on a DRY surface. When using electrical equipment, plugging in an electric cord etc

4. Never put conductive metal objects into energized equipment.

5. Always pick up and carry portable equipment by the handle and/or base. Carrying equipment by the cord damages the cord's insulation.

6. Unplug cords from electrical outlets by pulling on the plug instead of pulling on the cord.

7. Use extension cords temporarily. The cord should be appropriately rated for the job.

8. Use extension cords with 3 prong plugs to ensure that equipment is grounded.

9. Never remove the grounding post from a 3 prong plug so you can plug it into a 2 prong, wall Outlet or extension cord.

10. Re-route electrical cords or extension cords so they aren't run across the floor, under rugs or through doorways, etc. Stepping on, pinching or-rolling over a cord will break down the insulation and will create shock and fire hazards.

11. Don't overload extension cords, multi-outlet strips and wall outlets.

12. Heed the warning signs, barricades and/or guards that are posted when equipment or wiring is being repaired or installed or if electrical componel1ts are exposed.

B. Check for Unsafe Conditions (either before or while you're using equipment :)

1. Is the cord's insulation 'frayed, cracked or damaged, exposing the internal wiring?2. Are the plug's prongs bent, broken or missing, especially the third prong?
3. Is the plug or outlet blackened by arcing?
4. Was liquid spilled on or around the equipment?
5. Are any protective parts (or covers) broken, cracked or missing?
6. Do you feel a slight shock when you use the equipment?

7. Does the equipment or the cord overheat when it is running?
8. Does the equipment spark when it is plugged in or when switches or contr91s are used?

C. If you-observe any of these unsafe conditions:

1. Don't use (or stop using) the equipment.
2. Tag/label the equipment UNSAFE--DO NOT USE and describe the problem.
3. Notify your supervisor, Lab-Assistant or Lab Engineer as appropriate.

Electrical safety is for everyone because even contact with the standard low voltage electrical circuits which we constantly use, 'can be lethal.

College of Electrical & Mechanical

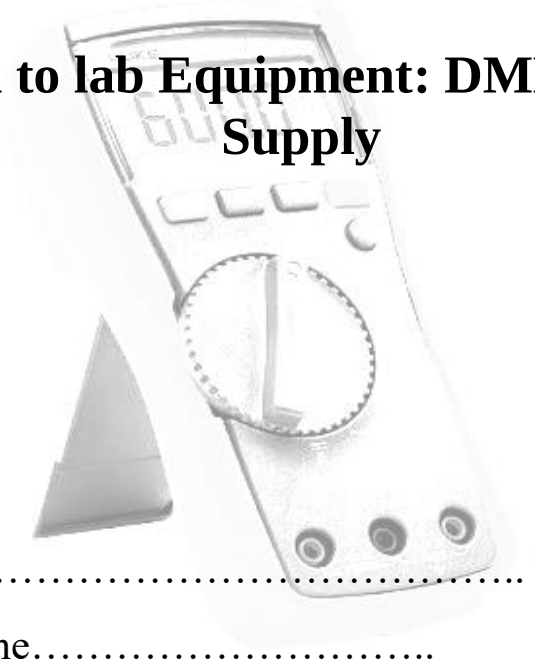
Engineering, NUST

Department of Basic Sciences & Humanities

Applied Physics

Lab No 1

Introduction to lab Equipment: DMM & DC Power Supply



Name……

Degree & Discipline……..

Group No…….

Date……

Date of Submission……

WORKING (60%)	PROBLEM UNDERSTANDING (20%)	RESULTS (20%)

Introduction to lab Equipment:

DMM & DC Power Supply

Objectives:

I. To learn the functions and working of DMM and DC Power Supply using Color-coded resistors.

Equipment Required-

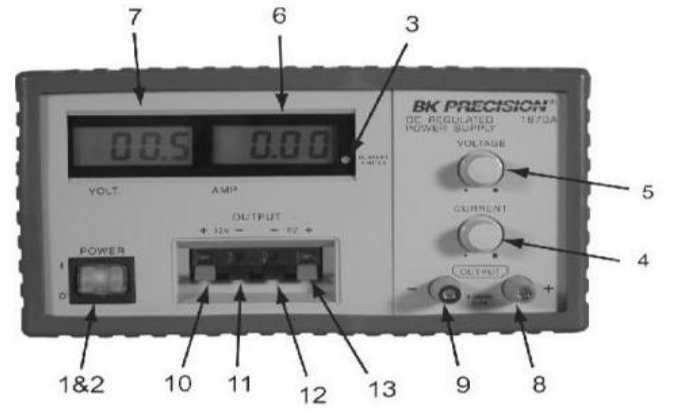
1. DMM (1x)
2. DC Power Supply(1x)
3. Breadboard (1x)
4. Color Coded Resistors(4x)
5. Connecting wire

INTRODUCTION

DC Power supply:-

The type of DC power supply discussed here derives its operating energy from the AC power line via an isolated winding on a transformer. A DC power supply is one that supplies a constant DC voltage to its output. The output can be fixed, variable or both. The basic components of a DC power supply are:

1. Power Transformer
2. Rectifier
3. Filter (usually a capacitor)
4. Voltage Regulator



Front Panel Description:-

1. POWER Switch.	Turns power supply on and off."I" : power switch ON "O" : power switch OFF Power ON light. Lights when power is turned on.

2. CURRENT LIMITED Indicator.	Lights when variable supply is operating in constant current mode.
3. CURRENT Control.	Adjusts maximum output current (0 to 5 Amps) of variable supply. Read value on A meter
4. VOLTAGE Control	Adjusts output variable voltage 0-30 V_{DC} voltage of variable supply. Read value on V Meter.
5. A Meter.	Indicates output current of variable supply.
6. V Meter	Indicates output voltage of variable supply.
7. OUTPUT + Terminal (Red).	Positive polarity output terminal.
8. OUTPUT - Terminal (Black).	Negative polarity variable output terminal.
9. 12V OUTPUT + Terminal (red).	Positive polarity 12 VDC output terminal.
10. 12V OUTPUT - Terminal (black).	Negative polarity 12 VDC output terminal.
11. 5V OUTPUT - Terminal (black).	Negative polarity 5 VDC output terminal
12. 5V OUTPUT + Terminal (red).	Positive polarity 5 VDC output terminal.

Digital Multimeter

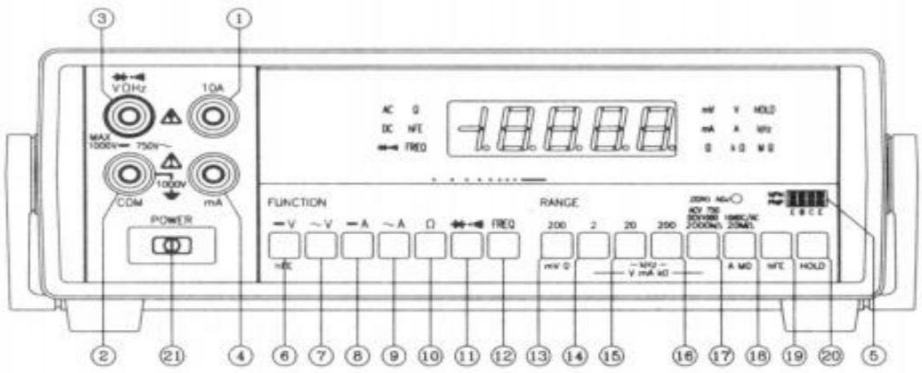


FIG 2-1. DM-441B FRONT VIEW

⑥	⎓ V	VOLTS DC/hFE	⑩	Ω	RESISTANCE
⑦	~V	VOLTS AC	⑪	⊣◁ •)))	DIODE AND CONTINUITY
⑧	⎓ A	AMPERES DC	⑫	FREQ	FREQUENCY
⑨	~A	AMPERES AC	㉑	⏻	POWER ON/OFF SWITCH

TABLE 2-3. FUNCTION SELECTED BY SWITCH

⑬ 200mV Ω	200mV,200Ω RANGE	⑰ 2000V mA k Ω	DC1000V/AC750V,2000mA,2000K Ω RANGE
⑭ 2V mA K Ω	2V,2mA,2K Ω RANGE	⑱ 10A M Ω	10A,20M Ω RANGE
⑮ 20V mA K Ω KHz	20V,20mA,20K Ω ,20KHz RANGE	⑲ hFE	hFE TEST
⑯ 200V mA k Ω KHz	200V,200mA,200K Ω ,200KHzRANGE	⑳ HOLD	DATA HOLD FUNCTION

TABLE 2-4. RANGE SELECTED BY SWITCH

Digital Multi Meter DMM:

A multimeter or a multitester, also known as a VOM (volt-ohm-milliammeter), is an electronic measuring instrument that combines several measurement functions in one unit. It can measure voltage, current, and resistance simply by moving dial.as shown in fig below. It can measure DC and RMS value. It can also be used to check continuity of circuit and one can easily check the fault in any circuit.



Resistance Color Chart:- Decoding the resistance from the color bands.

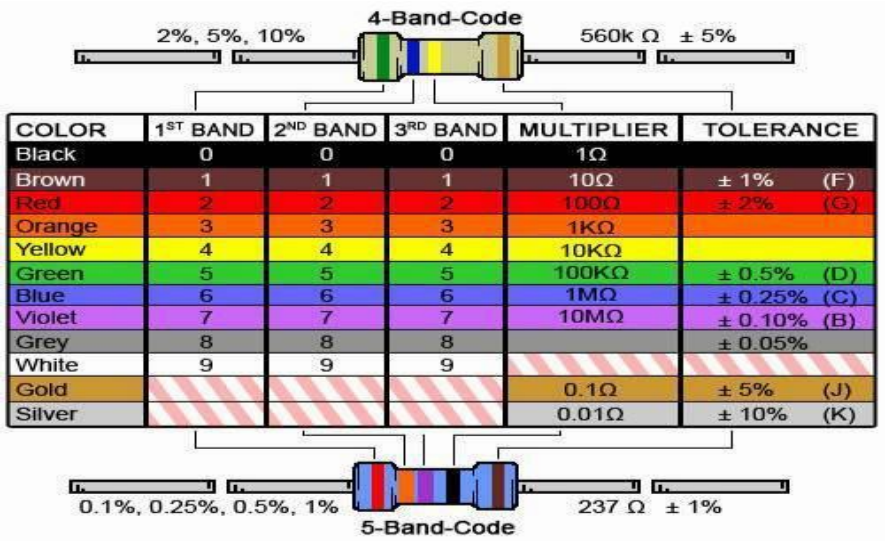
For standard precision resistors (four bands) : As mentioned above, **standard precision resistors use four color bands**. An image of the standard precision resistor is also shown above. For standard precision resistors,

a. The First band indicates the first digit of the resistance value.
b. The second band indicates the second digit of the resistance value.
c. The third band indicates the number of zeros to be added after the first two digits. Except when the color is silver or golden. Example 2 given below shows what should be done if the color is silver or golden.
d. The fourth band indicates tolerance.

Three band resistors: Resistors with three color bands can be decoded the same way as that of resistors with four color bands with the exception that the tolerance value is assumed to be 20%. Three bands indicate the nominal value of resistance and the tolerance is 20%.

For High precision resistors (five bands): High precision resistors uses five bands. The procedure to determine the resistance of high precision resistors (5 bands) resistors is quite similar to that of four band resistors. In 5 band resistors, the first three bands indicate the value, fourth band indicates multiplier and the fifth band indicates tolerance (usually ± 1%). Consider a resistor with color bands red-violet-black-silver-brown. The first three colors red (2), violet (7) and black (0) indicates the number 270. The fourth band-silver (0.01) indicates that we

should multiply 270 with 0.01, resulting in the nominal value of resistance to be 2.7 Ω. The fifth color (brown) indicates a tolerance of ± 1%.



COLOR	1ST BAND	2ND BAND	3RD BAND	MULTIPLIER	TOLERANCE
Black	0	0	0	1Ω	
Brown	1	1	1	10Ω	± 1% (F)
Red	2	2	2	100Ω	± 2% (G)
Orange	3	3	3	1KΩ	
Yellow	4	4	4	10KΩ	
Green	5	5	5	100KΩ	± 0.5% (D)
Blue	6	6	6	1MΩ	± 0.25% (C)
Violet	7	7	7	10MΩ	± 0.10% (B)
Grey	8	8	8		± 0.05%
White	9	9	9		
Gold				0.1Ω	± 5% (J)
Silver				0.01Ω	± 10% (K)

Part 1: Measurement of DC Power Supply Output Voltage Using DMM.

a. Connect DMM leads with 5V output of DC power supply, note down DMM value in table 1-1, and calculate percentage difference.
b. Connect DMM leads with 12V output of DC power supply, note down DMM value in table 1-1, and calculate percentage difference.
c. Connect DMM leads with output terminals of DC power supply and adjust 5 different voltages using variable knob on power supply. Note down voltages displayed on digital displays of power supply and DMM in table 1-1 and calculate corresponding % difference.

Part 2: Resistance Measurement:

a. Calculate resistance of each given resistor table shown in Fig and note down values in table 1-1.
b. Measure resistance sing DMM and note down vales in table 1-1.
c. Calculate % difference for each calculated and measured value and note down in Tab Using expression

$$\left|\frac{\text{measured} - \text{standard}}{\text{standard}}\right| \times 100$$

Observation and Calculation

Table 1-1

Sr. No.	Voltage displayed on DC Power Supply	Voltage measured by DMM	% Difference=
1			
2			
3			
4			
5			

Table 1-2

Sr. No.	Resistance Color Codes	Calculated Resistance with tolerance range	Measured Resistance (Using DMM)	% Difference
1				
2				
3				
4				
5				

Exercise

Short questions:

1. What is output voltage range of regulated output terminal of DC power supply?

   Ans……
   ……
   ……
   ……
   ……

2. What is the maximum current value that can be drawn from the fixed voltage terminals of the
   power supply?

   Ans……
   ……
   ……
   ……
   ……

3. What is the purpose of current limiting in power supply functions?

   Ans……
   ……
   ……
   ……
   ……

4. What CC and CV LED indicates?

   Ans……
   ……
   ……

……
……

5. What is purpose of silver and gold color codes?

Ans……
……
……
……
……

6. How 4 color & 5 color resistances are decoded? Give examples.

Ans……
……
……
……
……

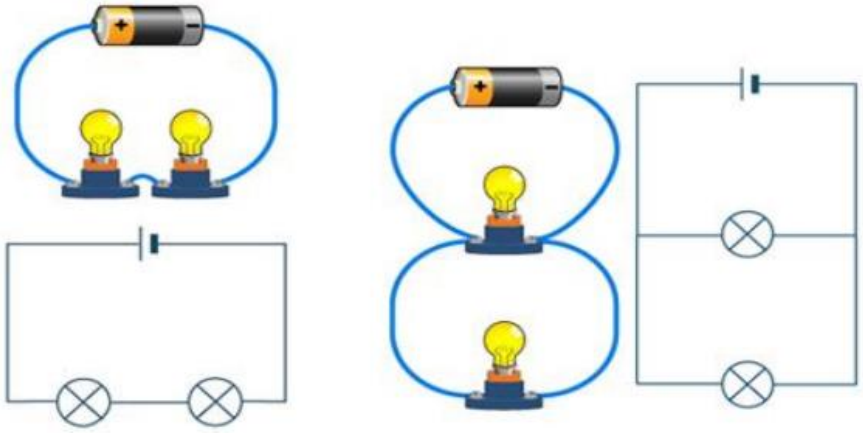
College of Electrical & Mechanical Engineering, NUST

Department of Basic Sciences & Humanities

Applied Physics

Lab No 2

Analysis of Series and Parallel Resistive Circuits



Series Circuit Parallel Circuit

Name……..

Degree & Discipline……..

Group No…….

Date……

Date of Submission……

WORKING (60%)	PROBLEM UNDERSTANDING (20%)	RESULTS (20%)

Analysis of Series and Parallel Resistive Circuits

Objectives:

I. To be able to develop simple series and parallel resistive circuits on breadboard.

II. Perform voltage and current measurements and verify results using Ohms Law.

Equipment Required:-

1. Breadboard
2. Assorted resistors
3. Connecting wires
4. DMM
5. Power supply
6. Leads

INTRODUCTION:
Theory KCL AND KVL:

According to KCL, the sum of currents entering anode is equal to sum of all currents leaving the node. Kirchhoff's Voltage Law or KVL, states that, the algebraic sum of all voltages within the loop must be equal to zero, the sum of voltage rise is equal to sum of voltage drops. This idea by Kirchhoff is known as the Conservation of Energy.

Series Circuit: In the first part of this experiment, we will study the properties of resistors, which are connected "in series". Figure.1 shows three resistors connected in series (a) and the equivalent circuit with the three resistors replaced by an equivalent single resistor (b). In a series circuit, the current through all circuit elements is same due to charge conservation and continuity of current. The voltage drop across each resistor is however determined by using Ohm's Law.

Applying KVL $V_o = V_1 + V_2 + V_3$

Using Ohm's Law $I_o R_{eq} = I_1 R_1 + I_2 R_2 + I_3 R_3$

Continuity of current $I_o = I_1 = I_2 = I_3$

Hence $R_{eq} = R_1 + R_2 + R_3$

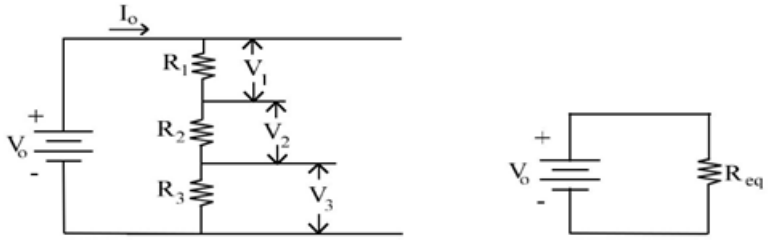


Figure 1(a) The series circuit (b) The equivalent circuit

Parallel Circuit: In the first part of this experiment, we will study the properties of resistors, which are connected "in parallel". Figure.2 shows three resistors connected in parallel (a) and the equivalent circuit with the three resistors replaced by an equivalent single resistor (b).

Applying KCL $I = I_1 + I_2 + I_3$

Using Ohm's Law $\frac{V}{R_{eq}} = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}$ where $V = V_1 = V_2 = V_3$

Hence $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

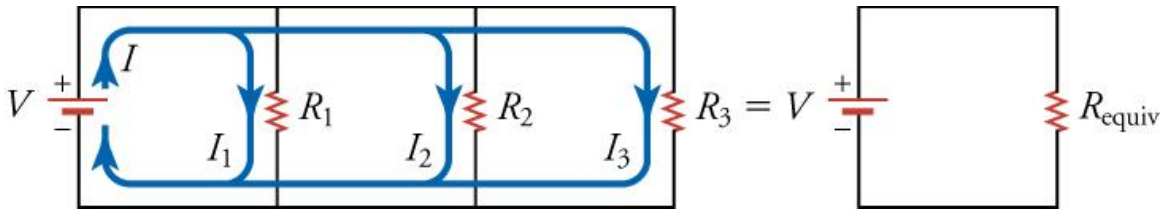


Figure 2 (a) The parallel circuit (b) The equivalent circuit

Voltmeter and Ammeter Connections with circuit:

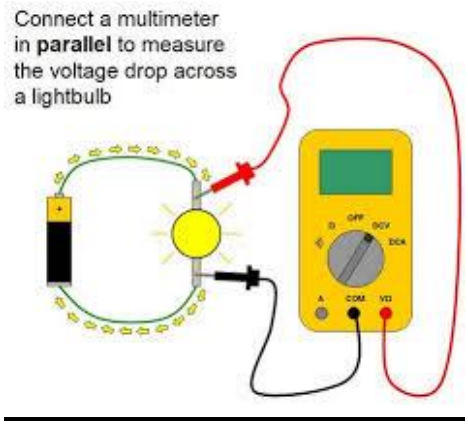


Figure 3 VOLTAGE MEASUREMENT

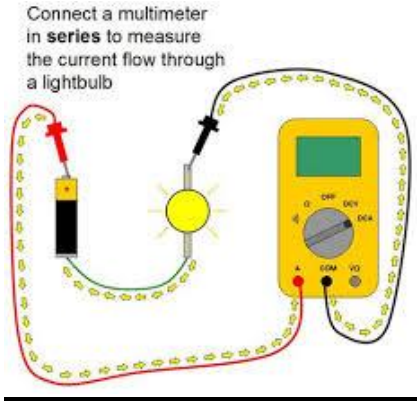


Figure 4 CURRENT MEASUREMENT

Observation and Calculation

Part 1: Resistors in Series:

Perform following measurements and calculations and note down values in Table 2-1

a. Decode each color-coded resistor.
b. Measure resistances using DMM.
c. Connect the resistors in series with source using a breadboard.
d. Set the source voltage at 5 V.
e. Measure voltage across each resistor.
f. Measure current through each resistor.
g. Measure equivalent resistance of resistor series combination.
h. Calculate the equivalent resistance (**Req**) of the circuit.
i. Perform calculations using fundamental laws.

$R_{eq} = R_1 + R_2 + R_3, \;\; I_S = {V_S}/{R_{eq}},$

Table 2-1

Sr. No.	Resistance No.	Resistance		Voltage		Current	
		Calculated (Using Color Code)	Measured (Using DMM)	Calculated $= I_S R_i$	Measured	Calculated $= I_S$	Measured
1	R_1						
2	R_2						
3	R_3						
4	R_{eq}						

Equivalent resistance in Series =

Part 2: Resistors in Parallel

Perform following measurements and calculations and note down values in Table 2-1

a. Decode each color-coded resistor.
b. Measure resistances using DMM.
c. Calculate the equivalent resistance (**Req**) of the circuit.
d. Connect the resistors in parallel with source using a breadboard.
e. Set the source voltage at 5 V.
f. Measure voltage across each resistor.
g. Measure current through each resistor.
h. Measure equivalent resistance of resistor series combination.
i. Perform calculations using fundamental laws.

$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$, $I_S = {V_S}/{R_{eq}}$

Table 2-2

Sr. No.	Resistance No.	Resistance		Voltage		Current	
		Calculated (Using Color Code)	Measured (Using DMM)	Calculated $= V_S$	Measured	Calculated $= \frac{V_1}{R_i}$	Measured
1	R_1						
2	R_2						
3	R_3						
4	R_{eq}						

Equivalent resistance in Parallel =

Exercise

Short questions:

1. Are the voltages V_1, V_2 and V_3 equal to each other? Why or why not?

Ans……
……
……
……
……..

2. Calculate the total voltage $V = V_1 + V_2 + V_3$. Explain why it has the value it does.

Ans……
……
……
……
……

3. Use Ohm's law to calculate the current through each resistor. (e.g., $V_1=I_1*R_1$, so $I_1=V_1/R_1$). For this calculation, use the measured value of the resistances. Record these calculated values in the table above. Is the result what you expected? Why?

Ans……
……
……
……
……

4. Is your measured value of R_{eq} similar to your calculated value? Explain.

Ans……
……
……
……
……

5. Are V_1 and V_2 equal to each other? Explain.

Ans……
……
……
……
……

6. Are I_1 and I_2 equal to each other? Explain.

   Ans……
   ……
   ……
   ……
   ……

7. Compare I_{total} to the I_1 and I_2. What do you notice?

   Ans……
   ……
   ……
   ……
   ……

8. Are the measured values of R_1 and R_2 equal to the values calculated using the color code chart? How much do they differ (calculate percent error)? Is this within the specified tolerance?

   Ans……
   ……
   ……
   ……
   ……

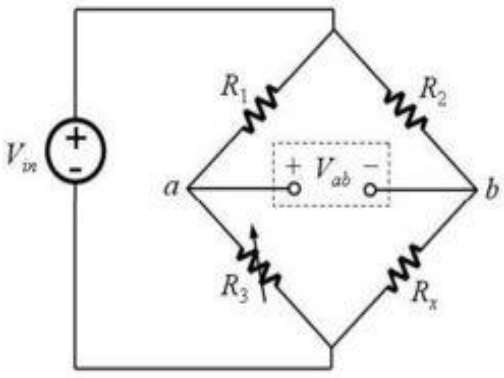
College of Electrical & Mechanical Engineering, NUST

Department of Basic Sciences & Humanities

Applied Physics

Lab No 3

Determination of Resistivity of unknown Material (Wire) using Wheatstone Bridge.



Name……..

Degree & Discipline……..

Group No…….

Date……

Date of Submission……

WORKING (60%)	PROBLEM UNDERSTANDING (20%)	RESULTS (20%)

Determination of Resistivity of unknown Material (Wire) using Wheatstone Bridge.

Objectives:

I. To learn how to measure diameter of wire

II. To study the use of slide wire bridge for precise measurements of resistances.

III. To determine the resistivity of a metal.

Equipment Required:-

1. Screw gauge(1x)
2. DMM(1x)
3. Resistance box(1x)
4. Slide wire bridge(1x)
5. wires to determine resistivity of wire(2x)
6. Galvanometer(1x)
7. Jockey(1x)
8. Connecting wires

INTRODUCTION:

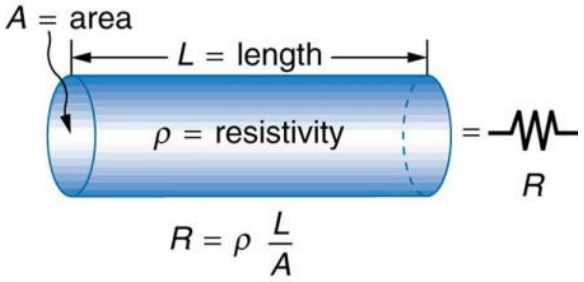
Theory:

The Electrical resistivity is the electrical resistance per unit length and per unit of cross-sectional area at a specified temperature. The SI unit of electrical resistivity is the ohm·metre (Ω·m). It is commonly represented by the Greek letter ρ, rho. It is an intrinsic property that quantifies how strongly a given material opposes the flow of electric current. A low resistivity indicates a material that readily allows the flow of electric current. Many resistors and conductors have a uniform cross section with a uniform flow of electric current, and are made of one material. In this case, the electrical resistivity ρ (Greek: rho) is defined as:

$$\rho = R\frac{A}{L} \qquad \text{(a)}$$

where

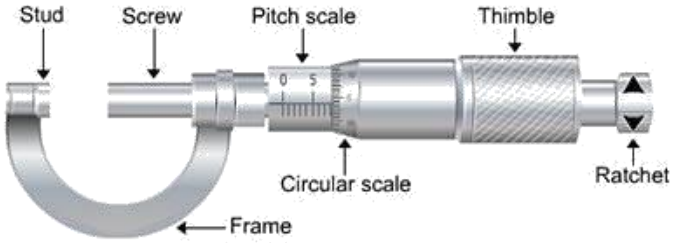
R is the electrical resistance of a uniform specimen of the material
L is the length of the piece of material
A is the cross-sectional area of the specimen



Screw Gauge:

The screw gauge is an instrument used for measuring accurately the diameter of a thin wire or the thickness of a sheet of metal. It consists of a U-shaped frame fitted with a screwed spindle which is attached to a thimble Parallel to the axis of the thimble; a scale graduated in mm is engraved. This is called pitch scale. A sleeve

is attached to the head of the screw. The head of the screw has a ratchet which avoids undue tightening of the screw. On the thimble there is a circular scale known as head scale which is divided into 50 or 100 equal parts. When the screw is worked, the sleeve moves over the pitch scale.A stud with a plane end surface called the anvil is fixed on the „U" frame exactly opposite to the tip of the screw. When the tip of the screw is in contact with the anvil, usually, the zero of the head scale coincides with the zero of the pitch scale.



Pitch of the Screw Gauge

The pitch of the screw is the distance moved by the spindle per revolution. To find this, the distance advanced by the head scale over the pitch scale for a definite number of complete rotation of the screw is determined. The pitch can be represented as;

$$\text{Pitch of the screw} = \frac{\text{Distance moved by screw}}{\text{No. of full rotations given}} \quad \text{.....(1)}$$

Least Count of the Screw Gauge

The Least count (LC) is the distance moved by the tip of the screw, when the screw is turned through 1 division of the head scale. The least count can be calculated using the formula;

$$\text{Least count} = \frac{\text{Pitch}}{\text{Total number of divisions on the circular scale}} \quad \text{.....(2)}$$

Zero Error and Zero Correction

To get the correct measurement, the zero error must be taken into account. For this purpose, the screw is rotated forward till the screw just touches the anvil and the edge of cap is on the zero mark of the pitch scale. The Screw gauge is held keeping the pitch scale vertical with its zero down wards.

When this is done, anyone of the following three situations can arise:

1. The zero mark of the circular scale comes on the reference line. In this case, the zero error and the zero correction, both are nil.
2. The zero mark of the circular scale remains above the reference line and does not cross it. In this case, the zero error is positive and the zero correction is negative depending on how many divisions it is above the reference line.

3. The zero mark of the head scale is below the reference line. In this case, the zero error is negative and the zero correction is positive depending on how many divisions it is below the reference line.



To find the diameter of the wire

With the wire shot between the screw and anvil, if the edge of the cap lies ahead of the Nth division of the linear scale. Then, linear scale reading (P.S.R.) = N.

If nth division of circular scale lies over reference line then, circular scale reading (H.S.R.) = n x (L.C.)

(L.C. is least count of screw gauge)

Total reading (T.R.) = P.S.R. + corrected H.S.R. = N + (n x L.C.). (3)

Wheatstone Bridge.

This is a device for precise measurements of resistances. It is based on the diagram in Fig. 1. At point A, an incoming current splits into two currents, Ix and I. In general, there could exist a "crossover current" between points C and D, which will be detected by the galvanometer. However, if the resistances R1 and R2 are properly chosen, the crossover current will be zero (as the galvanometer will show). This happens when the four resistances in Fig. 2 satisfy the relation Eq.4 This is known as the **BALANCED BRIDGE POSITION.**

$$\frac{R_1}{R_2} = \frac{R_x}{R_o} \qquad (4)$$

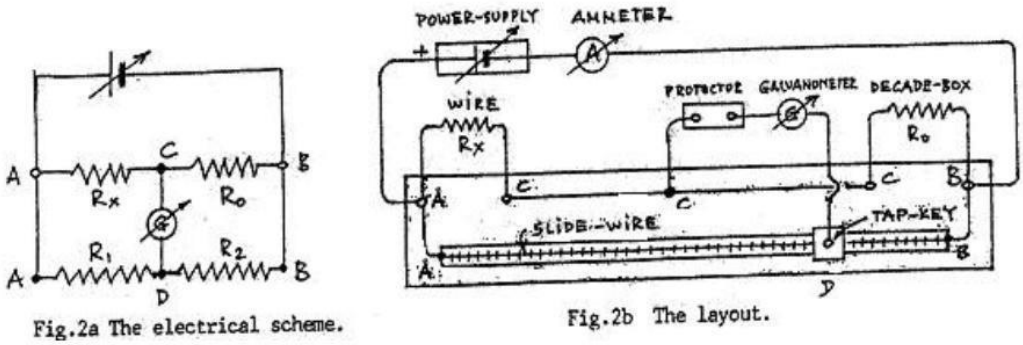


Fig.2a The electrical scheme. Fig.2b The layout.

In this experiment, the resistances R_1 and R_2 stem from two pieces of a uniform wire (same metal, same cross-section).Therefore, the ratio R1/ R2 is the same as the ratio L1/ L2 of their lengths. Then, if R_o is known, the unknown resistance Rx can be found from Eq 5.

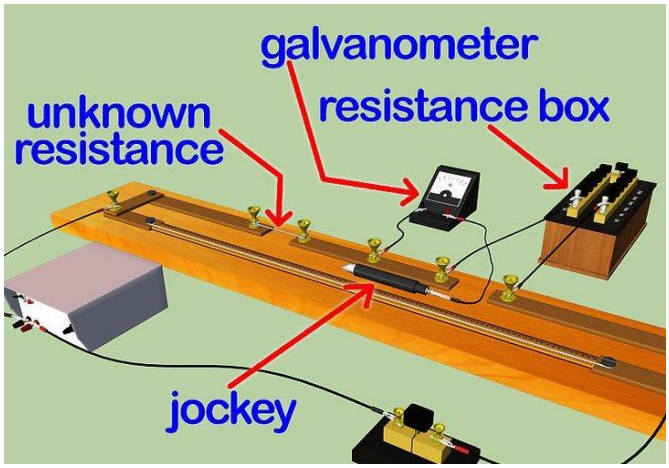
Procedure:-

Diameter Measurement:

1. Determine the pitch and least count of the screw gauge using the equations (1) and (2) respectively.
2. Bring the anvil, screw in contact with each other, and find the zero error. Do it three times and record them. If there is no zero error, then record "zero error nil"
3. Move the screw away from the anvil, place the wire, and move the screw towards the anvil using the ratchet head. Stop when the ratchet slips without moving the screw.
4. Note the number of divisions on the pitch scale that is visible and uncovered by the edge of the cap. The reading N is called the pitch scale reading(PSR)
5. Note the number (n) of the division of the circular scale lying over the reference line.
6. Find total reading using the equation 3 and apply zero correction in each case.

Resistance Measurement:-

1. For precise measurement of the resistance of sample wires, assemble the entire circuit according to the setup in Fig below. Make sure that all connection-screws are tight and that all connectors fit tightly in their sockets.
2. Measure the length of sample wire.
3. Extract suitable small resistance from fraction resistance box.
4. Tap Jokey on both sodes of slide wire.You should observe deflection on both sides of galvanometer at extreme ends of 100cm wire of slide Wire Bridge.
5. Move the tap-key back-and-forth and repeat the tapping, until you find a position where the galvanometer shows no noticeable deflection (if you cannot find such position, call your instructor). This position suggests BALANCED BRIDGE POSITION.
6. Record the values of R_0 and L_1, as well as L_2 = 100.0 cm - L1.
7. Repeat steps 3 to 6 for two different vales of Ro
8. Prepare the table 3-2:



Resistivity Measurement

Calculate the resistivity of each wire sample using relation (a) and data measured in previous Steps, recorded in table 3-3. Calculate the percentage difference for each wire.

Observation and Calculation

Measurements

Zero Error

1. Zero error…….mm
2. Zero Error……mm
3. Zero Error……mm

Mean ……mm

Mean Zero Correction……mm

Table 3-1Diameter Measurement

Object Placed	Pitch Scale	Head Scale Reading		Reading Diameter			% Diference
	Reading (N) mm	No of circular divisions on reference line (n)	[n x L.C] mm	Observed D0=N+n mm	Corrected D=D0 + c mm	Through Digital Screw gauge=D	
Copper wire							
Nichrome wire							

Table 3-2 Resistance Measurements

WIRE NO.	RUN #	Ro	L1	L2	Rx	Ave: < RX >
1(Copper)	1					
	2					
	3					
2(Nichrome)	1					
	2					
	3					

Table 3-3 Resistivity Calculations

Sr. No.	Wire Sample	Length (m)	Diameter (m)	Cross sectional Area $A=\pi r^2$ (m^2)	Rx (Ω)	Resistivity ρ (Ω m)	Theoretical Value (Ω m)	% Difference
1	Copper						1.72×10^{-8}	
2	Nichrome						1.10×10^{-6} to 1.50×10^{-6}	

Exercise

Short questions:

1. What is zero error of a screw gauge, and how we find it?

Ans……
……
……
……
……..

2. What is pitch of a screw gauge?

Ans……
……
……
……
……..

3. Derive Null Condition for Wheatstone Bridge

Ans……
……
……
……
……..

4. What should be the range of resistance values Ro extracted from Resistance box and why?

Ans……
……
……
……
……..

5. What is cross over current and when it is not zero?

Ans……
……
……
……
……..

College of Electrical & Mechanical

Engineering, NUST

Department of Basic Sciences & Humanities

Applied Physics

Lab No 4

Determine e/m ratio of electron using deflection method

Name……..

Degree & Discipline……..

Group No…….

Date……

Date of Submission……

WORKING (60%)	PROBLEM UNDERSTANDING (20%)	RESULTS (20%)

Determine e/m ratio of electron beam using deflection method

Objectives:

To measure experimentally e/m ratio of an electron and compare with theoretical value.

Equipment Required:

1. e/m apparatus
2. Scale

Introduction:

In this, experiment the forces exerted on electrons are due to electric and magnetic fields. The electrons are generated by thermionic emission. A beam of electrons is formed by accelerating electrons through a potential difference of ΔV Volts. The accelerated beam enters a magnetic field perpendicular to its direction of motion being produced by passing a magnetizing current through Helmoltz coil. Due to the presence of magnetic field, electrons experience a magnetic force, which deflect the electron beam forcing the electrons into a circular trajectory. The magnetic force acts as centripetal force. You can visualize this by using the right-hand-rule (remember the electrons have negative charge!).

By carefully noting the accelerating voltage, and the magnetizing current in the coils that create a magnetic field and radius of the circular path, one is able to make a precise measurement of the ratio of electron charge (e) and its mass (m).

Introduction of e/m ratio apparatus



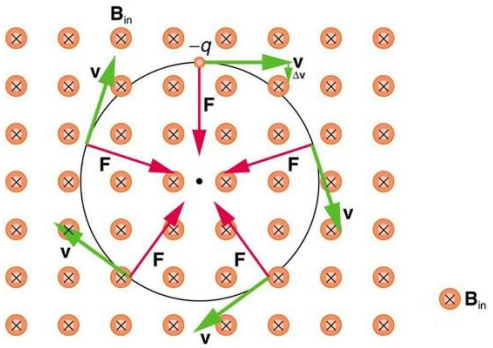


Figure 1 ELECTRON REVOLVING IN A CIRCULAR PATH IN A PERPENIDCULAR MAGNETIC FIELD IN CLOCKWISE DIRECTION

1. **Bainbridge Tube:**
   The Bainbridge tube is a fine-beam electron gun housed within a large spherical glass envelope. The electron gun directs an electron beam horizontally to the left. The envelope is filled with low-pressure mercury vapor, which emits a blue light along the path of electron beam. The gun is equipped with upper and lower deflection plates at its muzzle to demonstrate electrostatic deflection of the beam.
2. **Helmholtz coil:**
   The Helmholtz Coil is a pair of identical solenoid coils, aligned axially and separated by a distance equal to their radius, in this case 140mm. The coil windings are electrically connected in series-aiding. This particular arrangement produces a very uniform magnetic field with in the cylindrical region between the coils when an electrical current is passed through the coils. Each coil has a turn count of 150.

3. **Sliding Index:**
   The sliding index is used to sight the horizontal extremities of the electron beam, in order to measure the diameter of the beam. The top of the index is a V-notch along which the beam is to be sighted.

   Scale:
   The scale is used in conjunction with the sliding index (3) to measure the diameter of the deflected electron beam. It is secured to the apparatus by two clamps. The thumb screws of the clamps may be loosened and re-tightened in order to align the height of the scale with the horizontal diameter of the beam.

4. **Angular Scale:**
   The angular scale is clamped to the base of the Bainbridge tube (2), and is used to measure the axial rotation of the tube. The angular scale is secured by a tensioning screw, which may be loosened and retightened to adjust the position of the scale on the base.

   Fixed Index:
   The fixed index indicates the axial rotation of the Bainbridge (2) against the angular scale (5) on the base of the tube.

5. **Counter–Clockwise Magnetizing Current indicator:**
   The clockwise indicator illuminates to indicate that the direction of the current in Helmholtz coil is clockwise. Current direction is selected by the Magnetizing Current rocker switch (11).

   Clock Wise Magnetizing Current Indicator:
   The clock wise indicator illuminates to indicate that the direction of current in the Helmholtz coil is clockwise. Current direction is selected by the magnetizing current rocker switch (11).

   Deflection Voltage Dial:
   The Deflecting voltage dial is the control with which voltage across the tube's deflecting plates is adjusted. Approximate voltage is indicated by the pointer on the knob, against a scale printed on the front panel. This dial should be at full counter-clockwise during the e/m experiment.

Deflection Voltage polarity Switch:

The deflecting voltage polarity switch is three-position rocker switch with which polarity of deflecting voltage is selected upper plate positive in the upper position ;upper plate negative in the lower center [OFF] position during the e/m experiment.

6. **Accelerating Voltage Control Dial:**

The Accelerating voltage control dial is the control with which accelerating voltage to the tube'electron gun is adjusted, This voltage is indicated by the associated panel meter.

7. **Magnetizing Current Direction Switch:**

The magnetizing current direction switch is three-position rocker switch with which direction of conventional magnetizing current in the Helmholtz Coil (2) is selected: clockwise in the upper position; counter clockwise in the lower position; and current off in the center [OFF] position.

8. **Magnetizing Current Display:**
10. **Dark Box:**

The dark box encloses the Bain Bridge tube (1) and Helmholtz coil (2). The inside surfaces of the box are painted flat black to enhance visibility of the electron beam in the tube.

Theory of Operation:-

The Lorentz force is the magnetic force exerted on a moving electric charge by a magnetic field in which the charge is immersed this force is expressed in vector notation as

$$\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}$$

Where
F is the force in Newton,
q is the electric charge in coulombs,
v is the velocity in meters per second, and
B is the magnetic flux density in Webbers per square meter.

By vector cross-product, this force is perpendicular to both the charge velocity and the magnetic flux. If charge velocity and magnetic flux are perpendicular, all three vectors are mutually perpendicular; the Lorentz force then has magnitude

$$\boldsymbol{F} = qvB\sin(90) = qvB \ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

A force on an object perpendicular to the velocity of the object results in circular motion of the object so the Lorentz force in this case is centripetal force acting in opposition to a centrifugal force of magnitude

$$F = qvB = \frac{mv^2}{r} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

Where ,
m is the mass of the object in kilograms, and
r is the radius of the circular path in meters.

Equating these forces, and setting q=e (the charge on the electron) and m= (the mass of the electron), yields

$$evb = \frac{mv^2}{r} \text{.....(3)}$$

Dividing both sides of equation 3 by mvB yields

$$\frac{e}{m} = \frac{v^2}{vBr} = \frac{v}{rB} \text{.....(4)}$$

An electric charge e accelerated through a voltage ΔV attains a final kinetic energy equal to its initial potential energy eΔV. so that

$$\frac{1}{2}mv^2 = e\Delta V \text{ ..... (5)}$$

Solving for velocity yields

$$v = \sqrt{\frac{2e\Delta V}{m}} \text{ ..... (6)}$$

Using Eq4 and Eq6 give

$$\frac{e}{m} = \frac{2\Delta V}{r^2 B^2} \text{ ..... (7)}$$

By the Biot-Savat Law, the magnetic flux density at the axis of a solenoid coil (of which the Helmholtz coil is an instance) is

$$B = \frac{\mu_o}{5\sqrt{5}} \frac{IN}{R} \text{ ..... (8)}$$

Where,

μ_o is the permeability of the free space ($4\pi \times 10^{-7} c^{-2} Ns^2$),

I is the current through the coil winding in amperes,

N is the number of turn in the coil winding, and

R is the radius of the coil in meters.

Squaring equation 8, substituting into equation 7, and combining terms yields

$$\frac{e}{m} = \frac{125}{32\mu_o^2} \frac{\Delta V R^2}{r^2 I^2 N^2} \text{ ..... (9)}$$

Since voltage, radius of the coils, radius of the electron path, current through the coils and number of turns in the coils can all be measured; the charge-to-mass ratio of the electron can be determined by this expression.

The general procedure in this experiment will be to measure the radius of the electron beam path in the Bainbridge tube for a number of conditions of accelerating voltage and magnetizing current. The measured values of voltage, current, and radius for each condition; along with the number of turns in the coil and the coil radius; are substituted into equation 9. The results for all conditions are averaged to obtain a final value of the charge-to-mass ratio of the electron. The charge on the electron (as determined in the Millikan oil drop experiment, for example), can be divided by this ratio to determine the mass of a single electron.

Incidental Demonstrations

Component of charge Velocity Parallel to B: The socket for the Bainbridge is mounted to the chassis of the apparatus so that it can be rotated axially effect of rotating the tube is to "aim" the electron gun from side to side, thus imparting to the beam a component of velocity parallel to the magnetic flux. By vector cross-product, the Lorentz force on this parallel component of velocity is zero, so this component of the beam is not deflected. Meanwhile, the Lorentz force acts on the perpendicular component as it does in the e/m measurement, causing the beam to deflect in a circular path perpendicular to the flux.The superposition of these two motions is a helical path, with its axis parallel to the flux. Rotating the tube to various degrees changes the magnitude of the parallel component of velocity, thus changing the pitch of the helix.The base of the tube is fitted with an angular scale, and the chassis with a fixed index, to indicate the angle to which the tube is rotated.

Electrostatic Deflection of the Electron Beam:

An electron beam will also be deflected by an electric field, but in a different way as by a magnetic field. The electron gun in the Bainbridge tube is equipped with static deflection plates at its muzzle. When a voltage is impressed across these plates, an electric field is setup between them. As the electron beam passes between the plates and through this electric field, the beam is deflected by an angle. This angle increases with field intensity (controlled by the deflecting voltage); and decreases with beam velocity(controlled by accelerating voltage); Reversing the polarity of the voltage reverses the direction of deflection.Note: Magnetizing current should be turned off for this effect to best be seen. Don't permit the beam to strike any single point of the tube's glass envelope for an extended time.

Operation, Handling, and storage:

Although the Bainbridge tube will deliver reliable service over many hours of operation, it should still be considered a life-limited device. Following these guidelines will maximize tube life:

1. Allow the tube to warm up for a few minutes before applying accelerating voltage.
2. Bring accelerating voltage up slowly.
3. During operation of the tube, do not allow the electron beam to strike any one point on the tube for extended time which may result in puncture of tube.

4. When rotating the tube in the socket (or when removing the tube from the socket), grasp the tube by its Bakelite base, not by the glass neck.
5. Protect the tube from scratches and sharp impacts. If the apparatus is to be stored for an extended time, remove the tube from the apparatus and store it separately in its original packing box.

Additional operating Guidelines:

1. Adjust the magnetizing current to minimum before operating the magnetizing current direction switch.

2. Turn all controls to minimum and all switches to center-off before turning power on or off.

Protect the apparatus from vibration, shock and moisture. Install the protective cover over the front of the enclosure between uses. Store thee apparatus in a cool, dry location.

Initial SETUP:

Note Before applying power to the apparatus familiarize yourself with the operating precautions in Section. External meters may be connected to the rear-panel binding posts to measure accelerating voltage and magnetizing current, if higher precision is desired.

1. Set the Apparatus controls as follows: deflection voltage at minimum, deflection voltage polarity to "OFF", accelerating voltage at minimum, magnetizing current at minimum, magnetizing current direction to "OFF", Power to "OFF". Connect a power cord to the receptacle on the rear of the apparatus, and plug it into a 120V AC outlet.
2. Turn the apparatus on, Verify that the tube filament illuminates. Allow the cathode to heat up for a few minutes.
3. Slowly accelerating voltage up to 100 volts, at which point a continuous horizontal beam of electrons will suddenly emit from the electron gun.
4. Set the magnetizing current direction switch to "clockwise", and adjust current to about 1 ampere. This sets up a magnetic field between the coils, with flux parallel to the axis of the coils. Note that the electron beam, now immersed in the magnetic field, describes a circular path inside the tube. Rotate the tube in its socket as necessary to eliminate any axial component of velocity.

Measurement of the e/m ratio of the Electron:

1. Set accelerating voltage to 100 volts and magnetizing current to 1.0 amps as measured on the panel meters: record these values on your data sheet.
2. Position your eye directly in front of the left side of the beam's arc. Slide the index along the scale until its position is directly in line with the left side of the arc. It may be necessary to adjust the height of the scale. Record the index position on your data sheet. Repeat this step for the right side of the beam.
3. Repeat steps 1 and 2 with magnetizing currents of 1.25, 1.5, 1.75 and 2.0 Amps.
4. Repeat step 1 through 3 with acceleration voltages of 125, 150,175 and 200 volts.

5. Calculate the ration e/m for each combination of accelerating voltage and Magnetizing current in the data set. The radius of the beam is ½ the difference between the left and right index positions. Use a coil radius of 140mm, and a coil turn count of 150.
6. Average all calculated values to arrive at a final value of e/m. An example data set is shown in table. From the value of the charge on the electron, calculate the mass of the electron.

Observation and Calculation

Measurements:

Radius of Helmholtz coil=R=_____

Number of turns of Helmholtz coil=_____

Sr.N0	**ΔV** Accelerating Voltage **(**V)	**I** Magnetizing Current **(A**)	Beam diameter reading from left (cm)	Beam diameter reading from left (cm)	**r** Beam radius	**e/m ratio (observed)**	**e/m ratio (actual)**	**% error**
1	100	1.00						
2		1.50						
3		2.00						
4	150	1.00						
5		1.50						
6		2.00					1.75×10^{11}	
7	175	1.00						
8		1.50						
9		2.00						
10	200	1.00						
11		1.50						
12		2.00						

Exercise

Short questions:

1. Explain the purpose of accelerating Voltage and magnetizing current e/m ratio apparatus?

Ans……
……
……
……
……

2. What is effect of increasing accelerating voltage on beam radius?

Ans……
……
……
……
……

3. What is effect of increasing magnetizing current on beam radius?

Ans……
……
……
……
……

College of Electrical & Mechanical

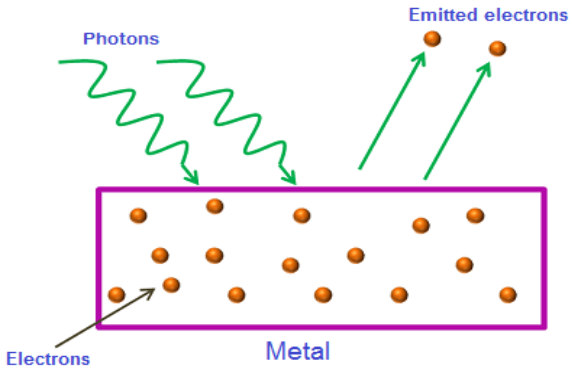
Engineering, NUST

Department of Basic Sciences & Humanities

Applied Physics

Lab No 5

Verification of inverse square law by studying variation of photoelectric current with intensity of light.



Name……..

Degree & Discipline……..

Group No……

Date……

Date of Submission……

WORKING (60%)	PROBLEM UNDERSTANDING (20%)	RESULTS (20%)

Verification of inverse square law by studying variation of photoelectric current with intensity of light.

5.1 Objective

1. To study variation of photo-electric current with intensity of light incident on photo-cell.
2. Verify inverse square law.

5.2 Apparatus:-

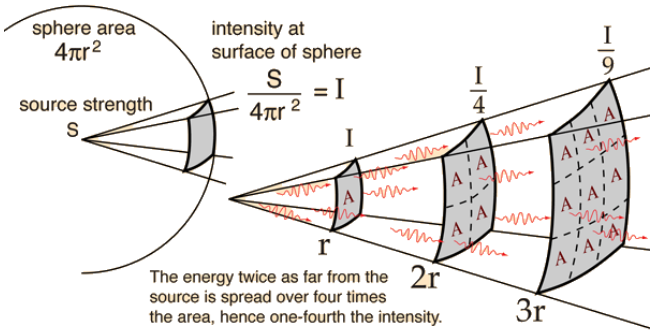
1. Wooden box with moving lamp inside the box and photocell at one end of the box
2. Microammeter

5.3 Introduction:-

5.3.1The Inverse Square Law

The inverse square law describes the intensity of light at different distances from a light source. Every light source is different, but the intensity changes in the same way. The intensity of light is inversely proportional to the square of the distance. This means that as the distance from a light source increases, the intensity of light is equal to a value multiplied by $1/d^2$,The proportional symbol, $\propto$, is used to show direct relationship.

Visible light is part of the electromagnetic spectrum, and the inverse square law is true for any other waves or rays on that spectrum, for example, radio waves, microwaves, infrared and ultraviolet light, x rays, and gamma rays. The intensity of visible light is measured in *candela* units, while the intensity of other waves is measured in Watts per meter squared (W/m^2).



5.3.2 The photoelectric effect:

Emission of electrons from the surface of matter including metals, gases, liquids and nonmetallic solids when light of a certain frequency is incident on it is known as photoelectric effect. In other words, it is the process of the removal of electrons from the surface of matter when rays of special frequency fall on the surface of matter. Because of the flow of these photoelectrons, the photoelectric current is produced.

5.4 Factors Affecting Photoelectric Effect:

Photoelectric current is produced as a result of photoelectric effect; therefore, understanding the factors which influence the photoelectric effect is very important. The previous studies on

photoelectric effect have presented the following factors which may have a direct impact on photoelectric effect.

1. **Intensity of Light:**

If a high intense light of frequency equal or greater than threshold frequency falls on the surface of matter, the photoelectric effect is caused. Since studying the impact of this factor is the focus of this research study, therefore, it would be discussed in detail however; one thing which is very clear is that the emission of electrons does not depend upon the intensity of light unless the frequency of light is greater than the threshold frequency.

2. **Frequency:**

If a beam of light with frequency equal to or greater than threshold frequency strike the surface of matter, photoelectric effect is produced. If frequency is less than the threshold frequency then photoelectric effect cannot be seen. The threshold frequency varies from matter to matter.

3. **Number of Photoelectrons:**

The increase in intensity of light increases the number of photoelectrons, provided the frequency is greater than threshold frequency. In short, the number of photoelectrons increases the photoelectric current.

4. **Kinetic Energy of Photoelectrons:**

The kinetic energy of photoelectrons increases when light of high energy falls on the surface of matter. When energy of light is equal to threshold energy then electrons are emitted from the surface whereas when energy is greater than threshold energy then photoelectric current is produced. The threshold frequency is not same for all kinds of matter and it varies from matter to matter.

5.5 Dependent and Independent Variables:-

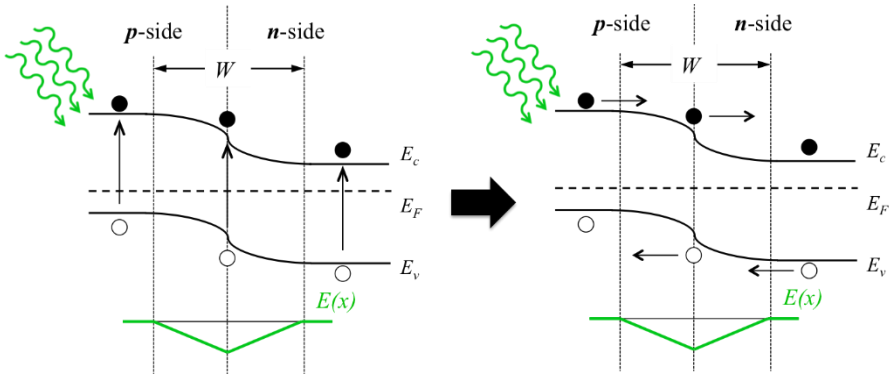
Basically the photoelectrons which move to produce photoelectric current are dependent upon different factors including the nature of material (In my experiment, the nature of cathode on which the light was falling), frequency of incident radiation, intensity of incident radiation and potential difference between the electrodes. In this experiment, we only consider one factor i.e. the intensity of incident light. The frequency of light was kept constant by only using the single source of light, while taking all the readings of the experiment. The potential difference of cathode was set before the conduction of the experiment and kept constant for all the readings. All readings were taken by performing the experiment on the same photocell, which ensured that the nature of the material used (cathode) was same for all the readings. The deflection of galvanometer is the dependent variable while the inverse of square of distance is the independent variable. All other factors affecting the photoelectric effect were kept constant.

Constant Variables

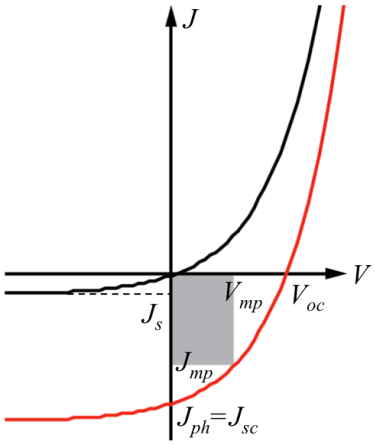
1- Air friction
2- Atmospheric pressure
3- Temperature of surrounding
4- Resistance

What happens when light with the energy above the bandgap shines onto the p-n junction diode?

Outside the depletion region, photo generated electrons can easily fall back onto valence band or "recombine" with their holes unless they diffuse into the depletion region. Remember that within the depletion region there is a built-in potential, or an electric field, which would immediately sweep the photogenerated holes and electrons into opposite directions. This means that the carriers generated within the depletion region or those that have diffused into the depletion region will be pushed by the field into the bulk of the material (holes into the ***p***-side and electrons into the ***n***-side).



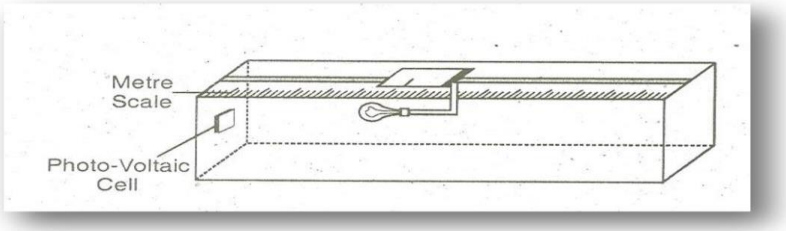
- When electron-hole pairs approach the junction they are pulled apart by the built-in field.
- Electrons are pushed into the n-side and holes are pushed into the p-side.
- If we connect p-side to n-side (i.e. make a short circuit), then these carriers will flow at zero applied voltage (just under built-in voltage). This means we will observe short-circuit current called photocurrent *Jph*.
- If we plot current-voltage characteristics for the p-n diode under illumination it will look shifted down by *Jph=Jsc*.

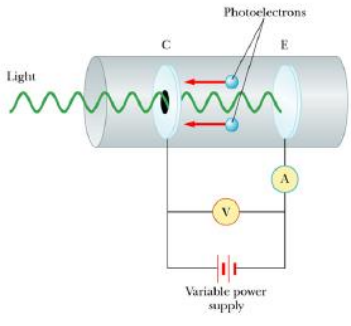


$$P_{\max} = J_{mp}V_{mp}$$

5.6 Procedure:

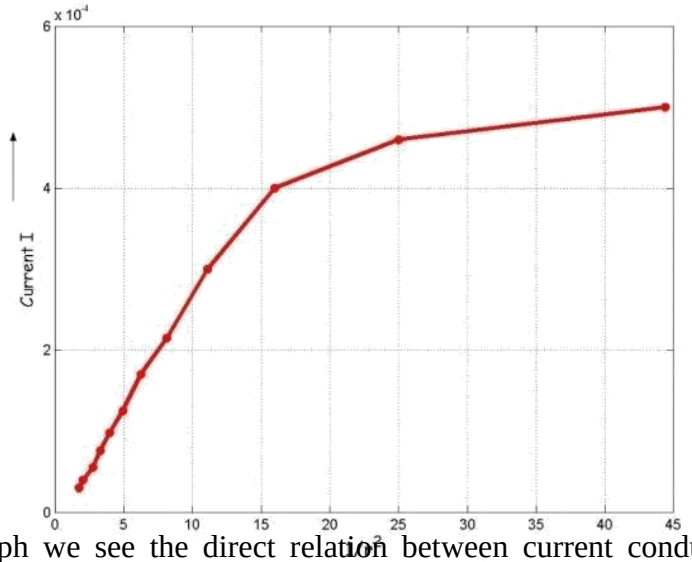
Switch on the lamp, slide the lamp, along the length of the box, increasing or decreasing the intensity of light on photocell. Observe the current at each specific distance of the lamp from the photocell and note down values in table 5-1.





A photocell reversed biased

5.7 Observations and graphical visualization of the data obtain



From the graph we see the direct relation between current conducted by the emission of electrons when light falls on the photocell and reciprocal of the distance between the light source and the photocell. In other terms Current I is directly proportional to the intensity of light because,

Intensity is proportional to $1/r^2$

The intensity, when increases, the chances of ejection of more electrons increases. The intensity only affects the number of electrons to be ejected that is why current is increasing with increasing intensity. The graph is not a perfect straight line, this may be due to some inaccuracy of the measuring instruments but up to an extent it shows the direct relation between the intensity of light and photoelectric current. In the graph, even for higher values of intensity the current increases gradually. This may be due to the fact that the maximum of the surface electrons from the cell are emitted. Now if the more energetic radiation falls upon the cell, the current probably would become directly proportional to intensity again. The energy of the radiation depends upon its frequency not on intensity. The radiation with high frequency would impart more energy to the electron and can penetrate deep inside also, to excite the electrons.

5.8 Precautions:-

1. Never expose the cell for a longer period of time to light.
2. Move the lamp gradually.
3. Take the measurements carefully.
4. The angle of incidence of light must not be changed on photocell

5.9 Observation and Calculation

Observed Data:-

Table 5-1:

Serial #	Distance of lamp from photo cell	Deflection of Galvanometer	$r^2(\pm 0.1)$	$1/r^2$
	'r' cm	Microampere (I)	cm^2	cm^{-2}
1.				
2.				
3.				
4.				
5.				
6.				
7.				
8.				
9.				
10.				
11.				
12.				
13.				
14.				
15.				

Exercise

Short questions:

1. What is photoelectric effect?

Ans……
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……
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……

2. What is inverse square law?

Ans……
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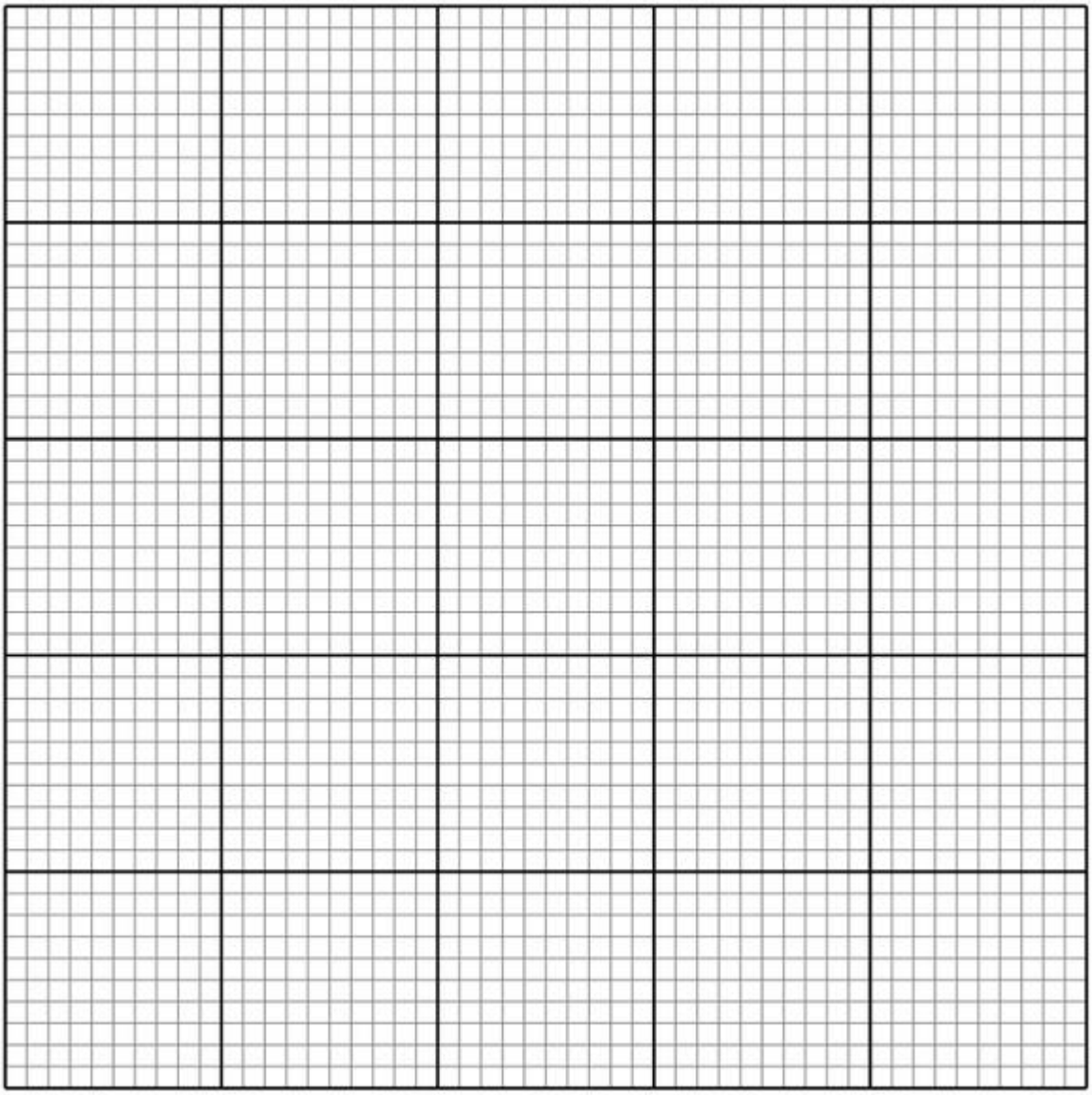
3. What is relationship between light intensity and current?

Ans……
……
……
……
……

4. What are different factors on which photocurrent depends on?

Ans……
……
……
……
……

APPLIED PHYSICA PHY-102



College of Electrical & Mechanical Engineering, NUST

Department of Basic Sciences & Humanities

Applied Physics

Lab No 6

Determination of the Planck's constant using a Photo Cell.

Name……..

Degree & Discipline……..

Group No…….

Date……

Date of Submission……

WORKING (60%)	PROBLEM UNDERSTANDING (20%)	RESULTS (20%)

Determination of the Planck's Constant using a Photo Cell.

Objectives:

1. To determine experimentally value of Planck's constant 'h' using Photocell
2. To calculate work function of material experimentally.

Apparatus:-

1. light source (12 V/35 W halogen tungsten lamp)
2. a dark-box which contains a vacuum phototube (sensitive component)
3. A DC amplifier, and five color filters Photocell

Precautions

1. Never expose the cell for a longer period of time to light
2. Move the lamp gradually.
3. Take the measurements carefully.
4. The angle of incidence of light must not be changed on photocell

Introduction:

In his 1901 paper, Planks stated that oscillators responsible for absorption and emission of light have discrete spectra of energy, and that the values in between are forbidden. In this model, the smallest unit of light that can be emitted or absorbed is

$$E = h\nu$$

where E is the energy of the `quantum', ν is the frequency of radiation, and h is a fundamental constant. The constant, later known as Planck's constant, has significance beyond the model of black body radiation, and is a major building block of Quantum Mechanics and Quantum Field Theory.

The Photoelectric Effect and Einstein's explanation

Photoelectric emission is a process in which light strikes a surface of material (e.g. a metal), and electrons come out. The kinetic energy of these electrons can be measured by subjecting them to a retarding electric field. The maximal kinetic energy is obtained when the electric field is strong enough to overcome all electrons.

In early 1900s, several experiments showed that:

1) The kinetic energy of electrons depended on the color of light (i.e., its frequency) and not on its intensity, and that dependence is linear. There exists some minimal frequency below which the light is unable to free any electrons, whereas above that critical frequency the light always succeeds in liberating electrons.
2) The photoelectric current (the number of electrons emitted) depended on intensity of the light.

Einstein took Planck's theory one step further and in 1905 stated that in the photoelectric process a photon of energy $E = h\nu$ is absorbed by electrons that are assumed to be bound within the surface of material with some energy W_o. The energy of the photon is used by the electron to escape the atom and the the rest is electron's kinetic energy $E_{K.E}$.

$$E = h\nu = w_o + E_{K.E}$$

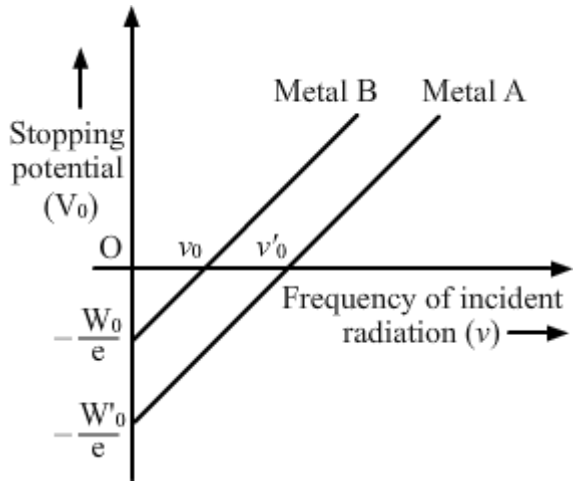
If a retarding potential V is used to stop electrons, that will happen when

$$eV = K.E\ .$$

(Note that in truth a third term exists - the kinetic energy of the recoiling material necessary to balance the electron's momentum, but it's neglected as infinitely small.) When solved for stopping potential V, this turns into

$$V = \frac{h}{e}\nu - \frac{w_o}{e}$$

so if we plot ***V*** as a function of frequency $\boldsymbol{\nu}$, we should get a linear dependence with a slope equal to ***h/e***! Here $\boldsymbol{W_0/e}$ is called *work function* and is a property of the material. This model thus makes a very definite prediction as to what the dependence of ***V*** on $\boldsymbol{\nu}$ ought to be. The goal of this experiment is to verify it, and, if true, to use it to measure ***h/e***.

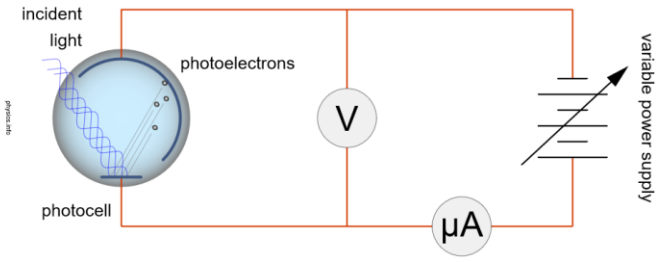


STOPPING POTENTIAL

Philip Nerad, assistant of Hertz, connected his photocell to a circuit with a variable power supply, voltmeter, and microammeter as shown in the schematic diagram below. He then illuminated the photoemissive surface with light of differing frequencies and intensities. Knocking electrons free from the photoemissive plate would give it a slight positive charge. Since the second plate was connected to the first by the wiring of the circuit, it too would become positive, which would then attract the photoelectrons floating freely through the vacuum where they would land and return back to the plate from which they started. Keep in mind that this experiment doesn't create electrons out of light, it just uses the energy in light to push electrons that are already there around the circuit. The photoelectric current generated by this means was quite small, but could be measured with the microammeter (a sensitive

galvanometer with a maximum deflection of only a few microamps). It also serves as a measure of the rate at which photoelectrons are leaving the surface of the photoemissive material.

Note how the power supply is wired into the circuit — with its negative end connected to the plate that isn't illuminated. This sets up a potential difference that tries to push the photoelectrons back into the photoemissive surface. When the power supply is set to a low voltage it traps the least energetic electrons, reducing the current through the microammeter. Increasing the voltage drives increasingly more energetic electrons back until finally none of them are able to leave the metal surface and the microammeter reads zero. The potential at which this occurs is called the *stopping potential*. It is a measure of the maximum kinetic energy of the electrons emitted as a result of the photoelectric effect**.**



Photocell connected to a circuit with a variable power supply, voltmeter, and micro ammeter as shown in the schematic diagram above.

What Lenard found was that the intensity of the incident light had no effect on the maximum kinetic energy of the photoelectrons (same value of stopping potential). Those ejected from exposure to a very bright light had the same energy as those ejected from exposure to a very dim light *of the same frequency*. In keeping with the law of conservation of energy, however, more electrons were ejected by a bright source than a dim source.

DESCRIPTION OF APPARATUS :

- Light Source : Halogen Lamp
- Light Intensity adjustor knob
- Voltage adjustor knob
- Voltage direction switch
- Current multiplier knob
- Voltage/current measuring (display mode) switch
- Photocell circuit fitted with DC power supply and a diode to avoid reverse current

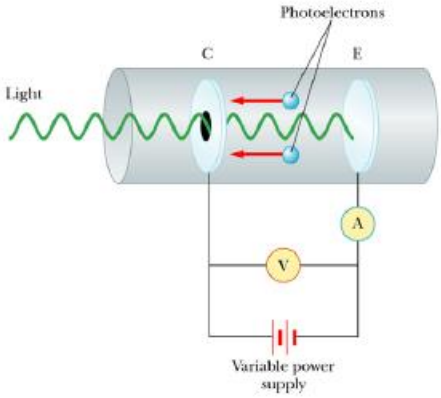


Figure 2 A reversed biased Photocell



Figure 1 Planck's constant apparatus

Determination of Planck's Constant

Step1: Use RED color filter. Rotate current multiplier knob to keep it at "x0.001" (to measure photocurrent up to three decimal places).

Step2: Keep voltage/current measuring (display mode) switch towards current and voltage direction switch toward –ve.

Step3: Rotate the voltage adjustor knob from zero such that photocurrent becomes equal to 0. Measure the corresponding voltage (known as stopping potential or cut off potential) by keeping voltage /current measuring (display mode) switch towards voltage.

Step4 : Repeat Steps 1, 2 and 3 for other given color filters.

Step5: Draw graph between stopping potentials and frequency for all colors. An, approximate straight line graph would come.

Step 6: Do the straight line (linear) fit of the plotted graph and get the slope of the line. The

slope will be equal to the Planck's constant.

Bonus !: The intercept on the Y axis will give you work function for the metal of the cathode of the photocell.

Observation and Calculation

Measurements:

Observed Data:-

Formula Used: The value of Planck's constant h is given by:

$$h = e(V_2 - V_1)\frac{\lambda_1\lambda_2}{c(\lambda_1 - \lambda_2)}$$

Where e = electronic charge =1.6 x 10^{-19}Coulombs

V2 = stopping potential for color2

V1 = stopping potential for color 1

c = velocity of light. = 3 x 10^8 m/sec

Sr. No. i	Color of filter	Wavelength(λi) of the filter	Frequency υi	Stopping Potential Vi
1	Red	635nm		
2	Orange	570nm		
3	Yellow	Yellow 540nm		
4	Green	Green 500nm		
5	Blue	Blue 460nm		

1- for red and green filters

$$h = e(V_2 - V_1)\frac{\lambda_1\lambda_2}{c(\lambda_1-\lambda_2)}$$ joules.sec

2- for green and blue filters:

$$h = e(V_2 - V_1)\frac{\lambda_1\lambda_2}{c(\lambda_1-\lambda_2)}$$ joules.sec

3. for yellow and blue filters:

$$h = e(V_2 - V_1)\frac{\lambda_1\lambda_2}{c(\lambda_1-\lambda_2)}$$ joules.sec

Mean value of Planck's constant = h = ….joule-sec.

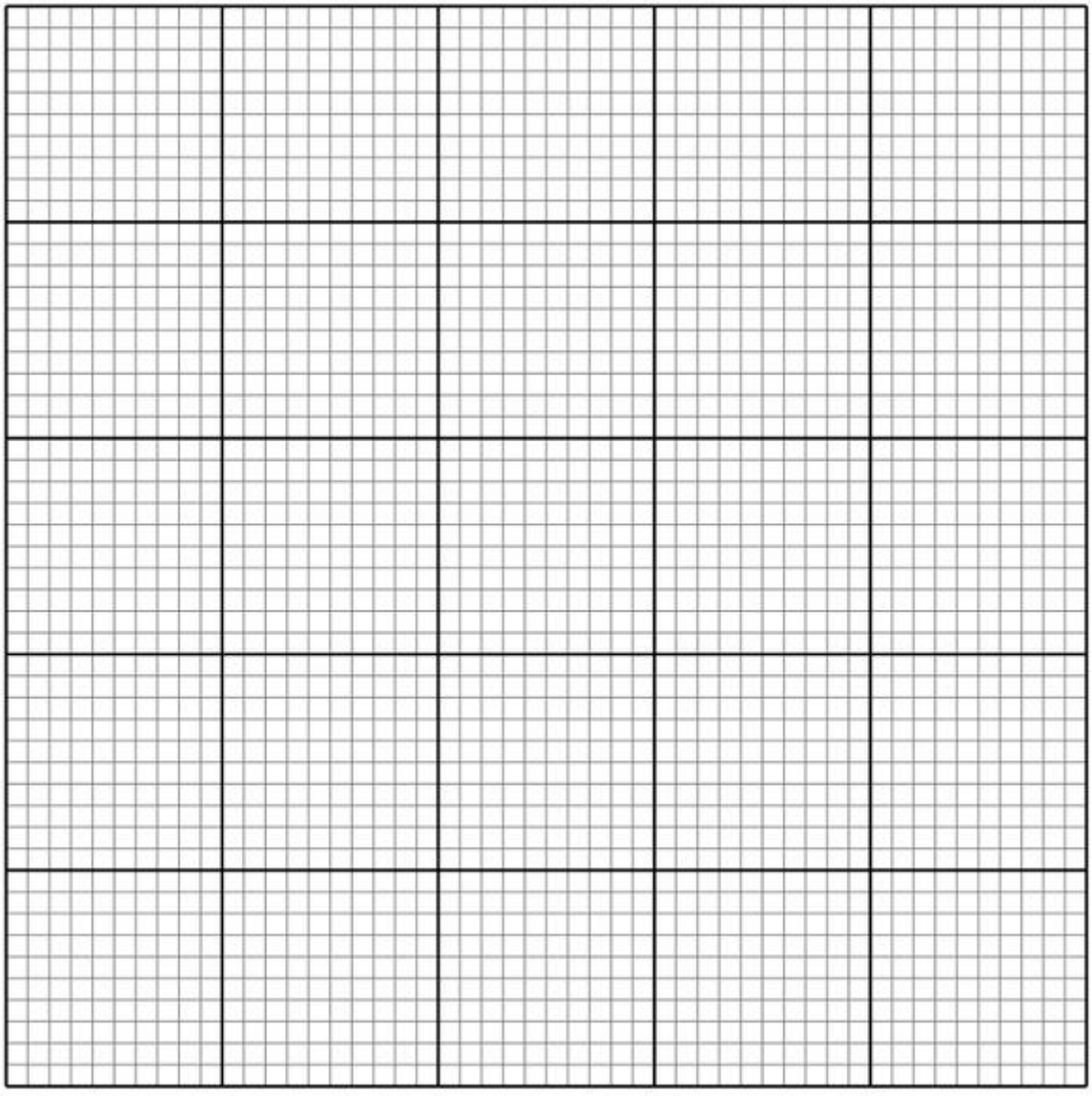
Standard Value: Standard value of Planck's constant = 6.625 *10^{-34} joule-sec

Percentage Error:

plot graph between frequency of incident light and corresponding stopping potential

Threshold Frequency of the material = _____

Work function of the material =_____



Exercise

Short questions:

1. Define each term in Einstein's equation for photoelectric effect.

Ans……
……
……
……
……..

2. Define Stopping Potential.

Ans……
……
……
……
……

3. Define Planck's constant. How it's used in photodiodes?

Ans……
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……
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……

4. What is a photocell?

Ans……
……
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……
……

5. Derive expression for the Planck's constant used in calculations.

Ans……IN REPORT……

College of Electrical & Mechanical

Engineering, NUST

Department of Basic Sciences & Humanities

Applied Physics

Lab No 7

Hook's Law: Determination of spring constant and effective mass of a spring by static and dynamic methods.

Name……..

Degree & Discipline……..

Group No…….

Date……

Date of Submission……

WORKING (60%)	PROBLEM UNDERSTANDING (20%)	RESULTS (20%)

Hook's Law: Determination of spring constant and effective mass of a spring by static and dynamic methods.

OBJECTIVE:

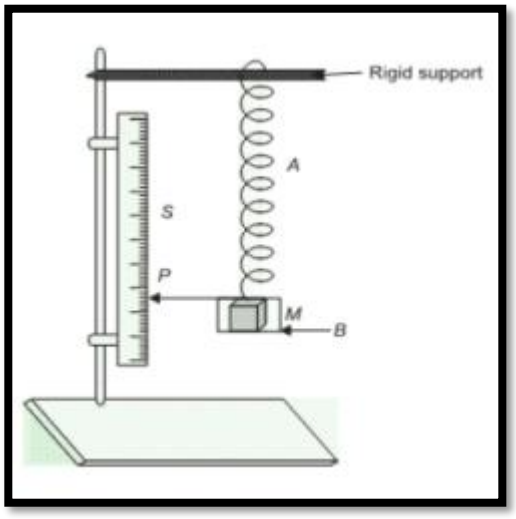
To determine the restoring force per unit extension of spiral spring by statical and dynamical methods and also to determine th mass of spring.

Apparatus used:

- spiral spring
- pointer
- Scale pan
- Slotted weights
- Stop watch

Theory:

A spiral spring consists of a uniform wire, shaped permanently to have the form of rectangular helix. A flat spiral spring is one in which the plane of spiral is perpendicular to the axis of cylinder. We deal here only with flat spiral spring. The spring is suspended from a rigid support with a hanger on the other end. A mass M is placed on the hanger such that the spiral spring is stretched vertically downward. If the extension of spring is small, the force from elastic reaction F is, from Hooke's law, proportional to the extension l, or



$$F \propto \Delta x$$

$$F = k\Delta x$$

Where k is constant giving measure of stiffness of spring and is called restoring force per unit extension of spring.But in equilibrium state

$$F = Mg$$

$$k\Delta x = Mg$$

$$k = Mg/\Delta x$$

If the mass at the end of spring is now displaced vertically downward and then released, then for small oscillations the restoring force at any instant is proportional to displacement Δx i.e. $F = -k\Delta x$

If $\frac{d^2x}{dt^2}$ is the acceleration of mass at that instant, the inertial reaction of system is $(M+m)\frac{d^2x}{dt^2}$,where m is the effective mass of the spring. Equating the sum of these forces to zero we get from Newton's third law of motion, the equation

$$(M+m)\frac{d^2x}{dt^2}+kx=0$$

Or

$$\frac{d^2x}{dt^2}+\frac{\text{kx}}{(\text{M+m})}=0$$

This equation represents a S.H.M whose period is given by

$$T=2\pi\sqrt{\frac{(M+m)}{k}}$$

If the experiment is performed with two masses M1 and M2 suspended successively at the end of the spring and the respective periods are T1 and T2, we have

$$T_1=2\pi\sqrt{\frac{(M_1+m)}{k}}$$

And

$$T_2=2\pi\sqrt{\frac{(M_2+m)}{k}}$$

Squaring eq.(2) and (3) and subtracting, we get

$${T_1}^2-{T_2}^2=\frac{4\pi^2(M_1-M_2)}{k}$$

$$k=\frac{4\pi^2(M_1-M_2)}{{T_1}^2-{T_2}^2}$$

From which k may be calculated.

Squaring eq.(2) and (3) and then dividing eq. (2) by Eq. (3) we have

$$\frac{T_1^2}{T_2^2}=\frac{M_1+m}{M_2+m}$$

Where on simplification

$$m = \frac{M_1 T_2^2 - M_2 T_1^2}{T_1^2 - T_2^2}$$

It can be shown that the mass of spring m' is three times the effective mass of spring.

Hence Mass of spring $$m' = 3[\frac{M_1 T_2^2 - M_2 T_1^2}{T_1^2 - T_2^2}]$$

$$T^2 = \left(\frac{4\pi^2}{k}\right) M + \left(\frac{4\pi^2}{k}\right) m$$

A plot of T^2 against M gives a straight line as shown in figure. The negative intercept OP on the x-axis equals M and may be used to find m graphically.

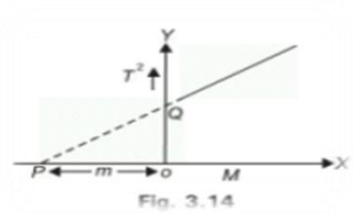


Fig. 3.14

Description of apparatus:

A spiral spring A whose restoring force per unit extension is to be determined is suspended from a rigid support as shown in figure 3.13. At the lower end of the spring, a small scale pan is fastened. A small horizontal pointer P is also attached to the scale pan. A millimeter scale S is also set in front of the spring in such a way that when spring vibrates up and down, the pointer freely moves over the scale.

Procedure: (a) Statistical Method

1. Hang a spiral spring A from a rigid support as shown in figure 3.13 and attached a scale pan B.
2. With no load in the scale-pan, note down the zero reading of the pointer on the scale.
3. Place gently in the pan a load of, say 100 gm.
4. Now the spring slightly stretches and the pointer moves down on the scale. In the steady position, note down the reading of the pointer. The difference of the two readings is the extension of the spring for the load in the pan.
5. Increase the lead in the pan in equal steps until maximum permissible load is reached and note down the corresponding pointer readings on the scale.
6. The experiment is repeated with decreasing loads.
7. Plot a graph as illustrated in figure between the load and the scale readings taking the load on X-axis and corresponding scale readings on Y-axis. The graph will be a straight line. Measure PR and QR and calculate k from the formula
   k= QR/PR*G Newton/meter.
8. **Observation:** Mass of the pan =

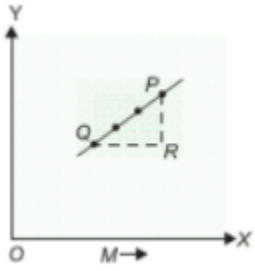


Fig. 3.15

Calculations: statistical experiment

The restoring force per unit extension of the spring

$$k = \frac{mg}{\Delta x} = \cdots Newton/meter$$

A graph is plotted between the load and scale readings described in point (7) in the procedure. From graph

PR= …….

QR= ……...

$$k = \frac{QR \times g}{PR} = \cdots Newton/meter$$

Result: The restoring force per unit extension of the spring= ……. Newton/meter

The mass of the spring = …… gm

Sources of errors and precautions

1. The axis of the spiral spring must be vertical.
2. The scale should be set up vertically and should be arranged to give almost the maximum extension allowed.
3. The pointer should move freely over the scale and should be just not in contact with it.
4. The spiral spring should not be stretched beyond elastic limit.
5. The load in the scale-pan should be placed gently and should be increased in equal steps.
6. While calculating the mean extension of the spring for a certain load, successive difference between consecutive readings of the pointer on the scale should not be taken.

Procedure: (b) dynamical Method

1. Load the pan. Displace the pan vertically downward through a small distance and release it. The spring performs simple harmonic oscillations.
2. With the help of stop watch, note down the time of a number of oscillations (say 20 or 30). Now divide the total time by the number of oscillations to find the time period (time for one oscillation) T_1.
3. Increase the load in the pan to M_2. As described above, find the time period T_2 for this load.
4. Repeat the experiment with different values of load.
5. Plot a graph between T^2 and m. The slope is $\frac{4\pi^2}{k}$ and the intercept on the negative side of the x-axis is m.

Observations: Measurement of periods T_1 and T_2 for the loads M_1 and M_2 . Least count of stop-watch=sec

Calculations: Restoring force per unit extension of spring

$$k = \frac{4\pi^2(M_1 - M_2)}{{T_1}^2 - {T_2}^2}$$

=…..Newton/metre

Mass of spring $$m' = 3\left[\frac{M_1 T_2^2 - M_2 T_1^2}{T_1^2 - T_2^2}\right]$$

=……gm

Result: The restoring force per unit extension of spring as determined by dynamical experiment=…….Newton/m

The mass of spring=……..gm

Sources of error and precautions

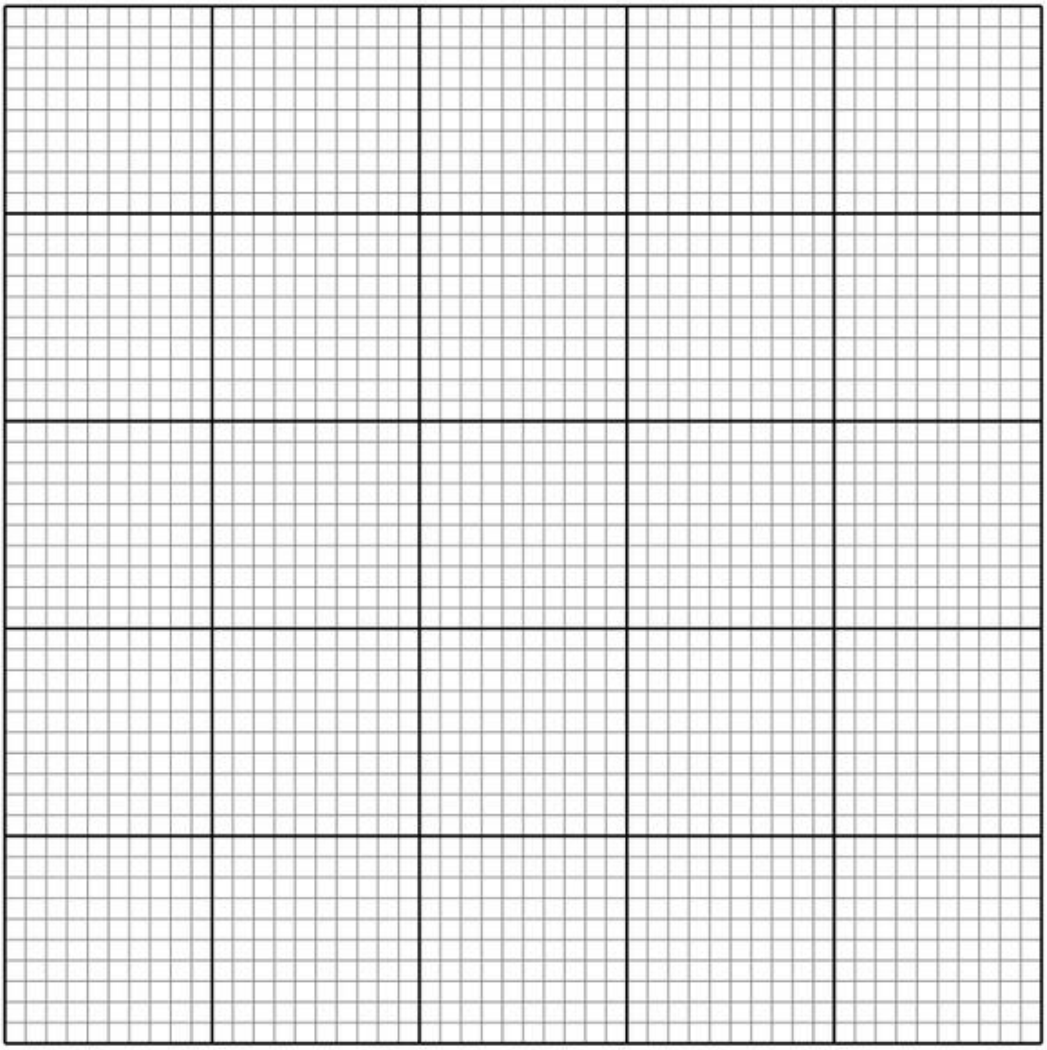
1. The spiral spring should oscillate vertically.
2. The amplitude of oscillation should be small.
3. As in the expression for k and m' the periods T_1 and T_2 occur raised to second power they should be accurately determined by timing a large number of oscillations correct upto the value measureable with the stop-watch.

Observation and Calculation

Table 7-1 Static Method:

Initial reading with zero load=X0=_____

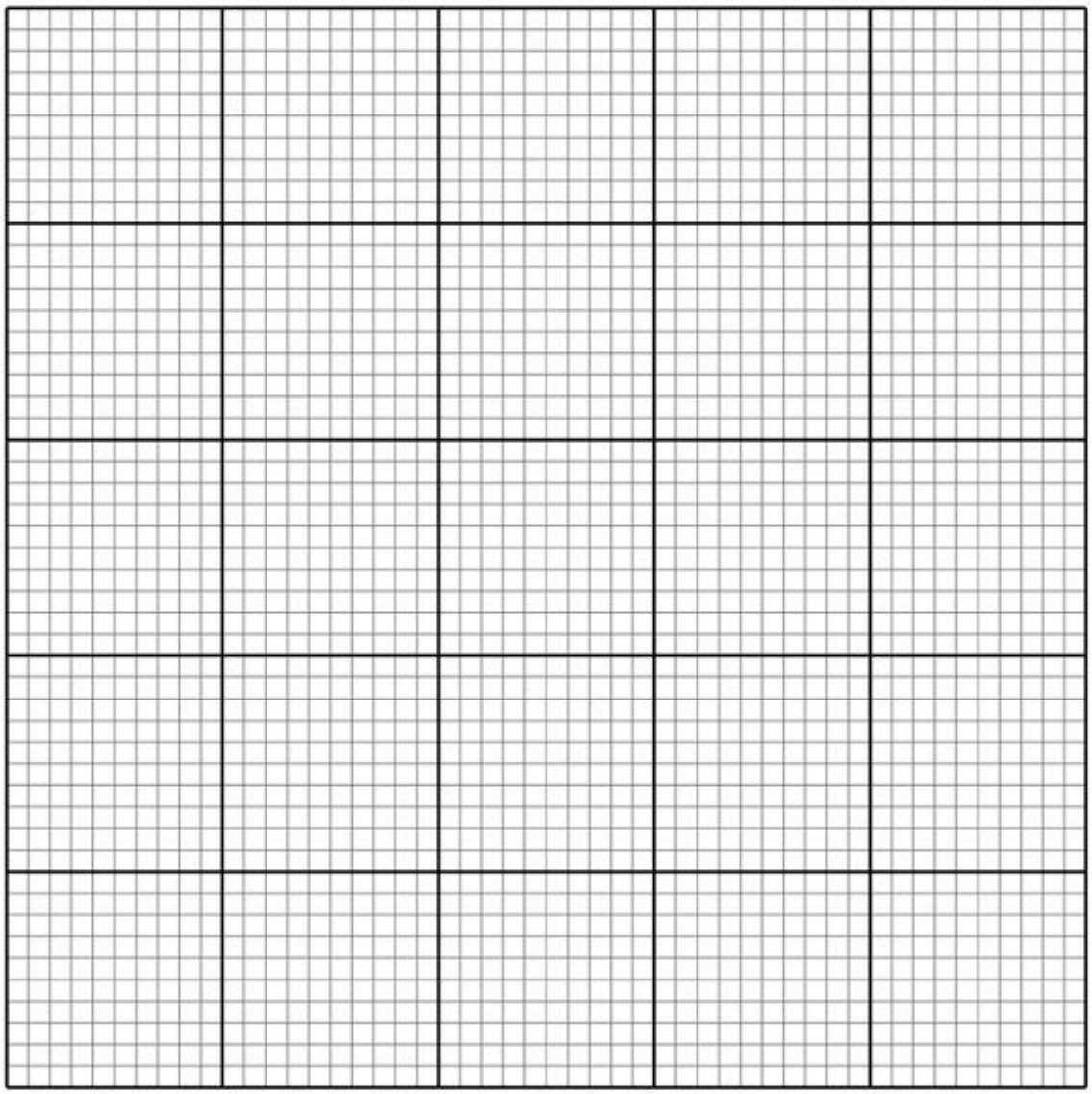
Sr.No.	m (kg)	Reading of pointer on the scale X (cm)			Δx (m)=X-X0
		Load increasing	Load decreasing	Mean	
1					
2					
3					
4					
5					



Observation and Calculation

Table 7-2 Dynamic Method

Sr. No.	m kg	10 T(s)	T(s)	$T^2(s^2)$
1				
2				
3				
4				
5				



Exercise

Short questions:

1. What is difference between static and dynamic method?

Ans……
……
……
……
……..

2. Define effective mass of the spring.

Ans……
……
……
……
……

3. Define spring constant.

Ans……
……
……
……
……

APPENDIX:-

EFFECTIVE MASS OF AN IDEAL UNIFORM SPRING,

Vertical spring-mass system

The effective mass of the spring in a spring-mass system when using an ideal spring of uniform linear density is 1/3 of the mass of the spring and is independent of the direction of the spring-mass system (i.e., horizontal, vertical, and oblique systems all have the same effective mass). This is because external acceleration does not affect the period of motion around the equilibrium point.

We can find the effective mass of the spring by finding its kinetic energy. This requires adding all the mass elements' kinetic energy, and requires the following integral, where u is the velocity of mass element

$$T = \int_m \frac{1}{2}u^2\, dm$$

Since the spring is uniform, $dm = \left(\frac{dy}{L}\right) m$, where L is the length of the spring. Hence,

$$T = \int_0^L \frac{1}{2}u^2 \left(\frac{dy}{L}\right) m$$
$$= \frac{1}{2}\frac{m}{L}\int_0^L u^2\, dy$$

The velocity of each mass element of the spring is directly proportional to its length, i.e. $u = \frac{vy}{L}$, from which it follows:

$$T = \frac{1}{2}\frac{m}{L}\int_0^L \left(\frac{vy}{L}\right)^2 dy$$
$$= \frac{1}{2}\frac{m}{L^3}v^2\int_0^L y^2\, dy$$
$$= \frac{1}{2}\frac{m}{L^3}v^2\left[\frac{y^3}{3}\right]_0^L$$
$$= \frac{1}{2}\frac{m}{3}v^2$$

Comparing to the expected original kinetic energy formula $KE = \frac{1}{2}mu^2$

We can conclude that effective mass of spring in this case is $\frac{m_S}{3}$

Reference:- https://en.wikipedia.org/wiki/Effective_mass_(spring%E2%80%93mass_system)



College of Electrical & Mechanical

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Applied Physics

Lab No 8

Compound Pendulum: Determination of radius of gyration K and acceleration due to gravity g.

Name……..

Degree & Discipline……..

Group No…….

Date……

Date of Submission……

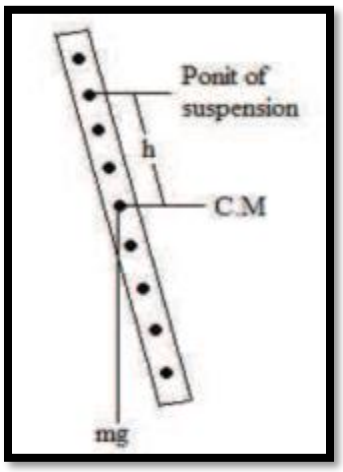
WORKING (60%)	PROBLEM UNDERSTANDING (20%)	RESULTS (20%)

Compound Pendulum: Determination of radius of gyration K and acceleration due to gravity g.

OBJECTIVE:

Use the compound pendulum to find:

1) The acceleration due to gravity g.
2) The moment of inertia of the rod.



THEORY:

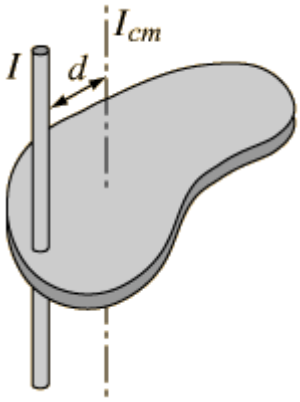
Any extended object mounted on a horizontal axis so as to oscillate under the force of gravity is a compound pendulum. The one used in this experiment is a uniform rod suspended at different locations along its length. The period T of a compound pendulum is given by

$$T = 2\pi\sqrt{\frac{I}{Mgh}} \quad (1)$$

Where:

I is the rotational inertia of pendulum about the axis of suspension. M is the pendulum mass and h is the distance between the suspension point and the center of mass. Using the parallel axis theorem

$$I = I_G + Mh^2 \quad (2)$$



I_G is the rotational inertia of body about its center of mass and it is given by

$$I_G = MK^2 \quad (3)$$

Substituting eq (3) in eq (2)

$$I = M(h^2 + K^2) \quad (4)$$

Where K is the radius of gyration, length of equivalent simple pendulum such that both have same moment of inertia and h is the displacement of parallel axis of rotation with respect to the axis passing through center of mass of the object. Substituting eq.(4) in eq (1)

$$T = 2\pi\sqrt{\frac{h^2 + K^2}{gh}} \quad (5)$$

The period of the simple pendulum is given by

$$T = 2\pi\sqrt{\frac{L}{g}} \qquad (6)$$

The period of a compound pendulum equals the period of a simple pendulum of a length

$$L = \frac{h^2 + K^2}{h} \qquad (7)$$

This equation can be solved to find L and K:

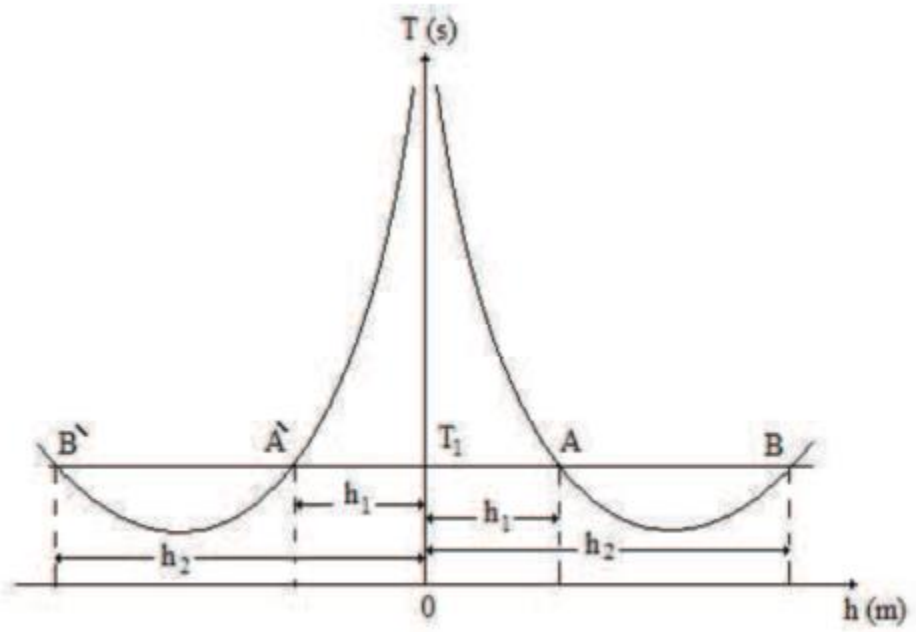
$$L = h_1 + h_2 \qquad (8)$$

$$K = \sqrt{h_1 h_2} \qquad (9)$$

PROCEDURE:

1. First hang the pendulum horizontally and move it until it reaches equilibrium so you can find the center of mass and mark it.
2. Secondly hang it vertically inserting the tip the knife in the first hole from the center of mass. Then set it oscillating through a small angle.
3. Measure the time needed for 20 oscillations and corresponding h.
4. Repeat step 2 and 3 for the other holes.
5. Record your measurements in a table.

GRAPH:



Observation and Calculation

MEASURMENTS AND RESULTS:

M=

One side from C.M			The other side of C.M		
h(m)	20T(s)	T(s)	h(m)	20T(s)	T(s)

From the graph:

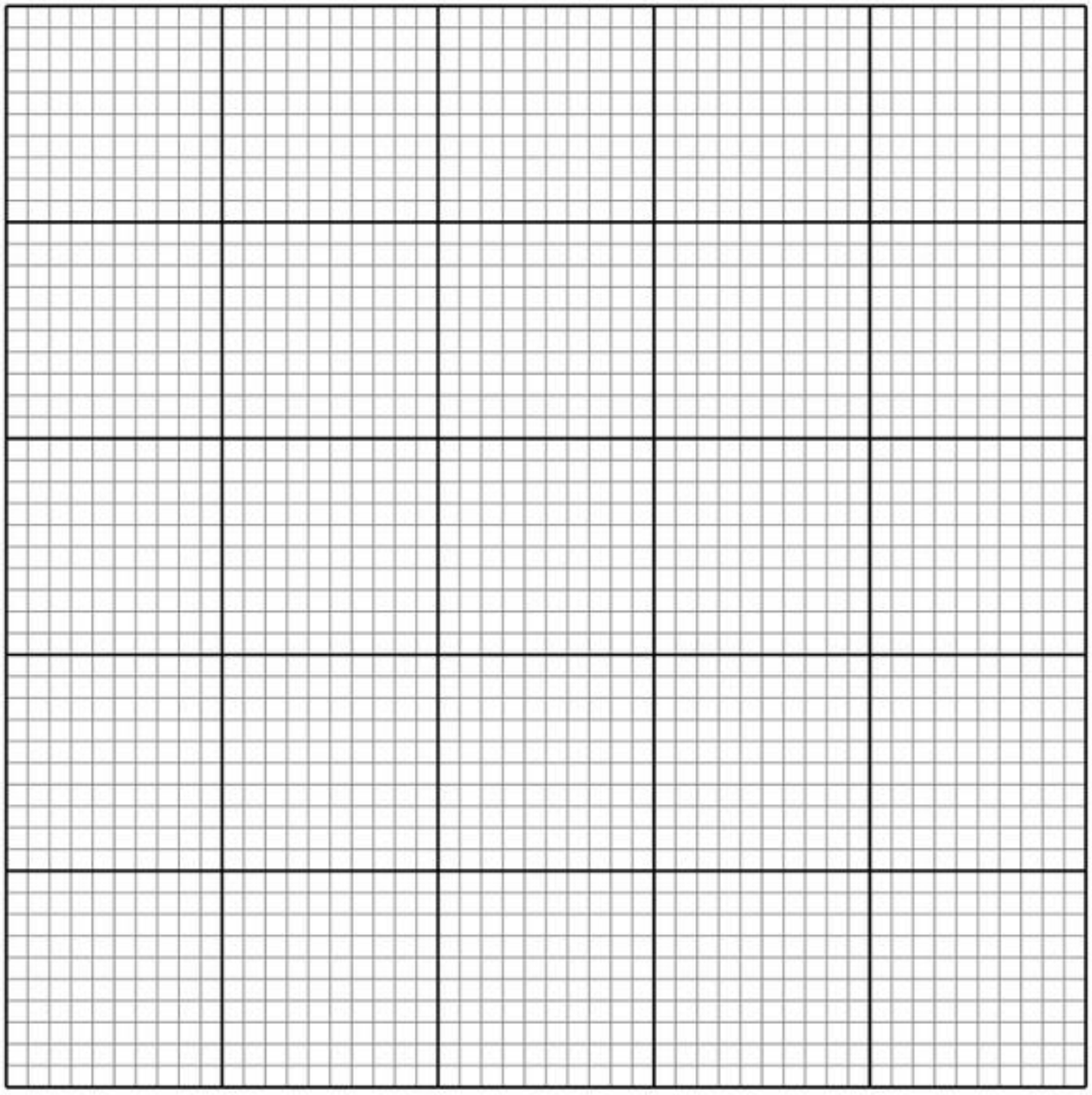
T	h_1(average)	h_2(average)	$L=h_1+h_2$	$K = \sqrt{h_1 h_2}$	$g = \frac{4\pi^2 L}{T^2}$	$I_G = MK^2$
(s)	(m)	(m)	(m)	(m)	(m/s^2)	Kgm^2

Finally calculate:

g(average)

I_G (average)

Results :



Exercise

Short questions:

1. What is difference between simple and compound pendulum?

Ans……
……
……
……
……..

2. Define moment of inertia.

Ans……
……
……
……
……

3. Define radius of gyration.

Ans……
……
……
……
……

4. Define parallel axis theorem.

Ans……
……
……
……
……
……

5. Solve quadratic equation given by Eq.7 for L and K as given by Eq8 and Eq9.

Ans……IN REPORT……

College of Electrical & Mechanical

Engineering, NUST

Department of Basic Sciences & Humanities

Applied Physics

Lab No 9

Introduction to Function generator and Oscilloscope

Name……..

Degree & Discipline……..

Group No…….

Date……

Date of Submission……

WORKING (60%)	PROBLEM UNDERSTANDING (20%)	RESULTS (20%)

Introduction to Function generator and Oscilloscope

Objective: To familiarize the students with basic operation of digital oscilloscope and audio generator.

Equipment Required:-

Audio Generator

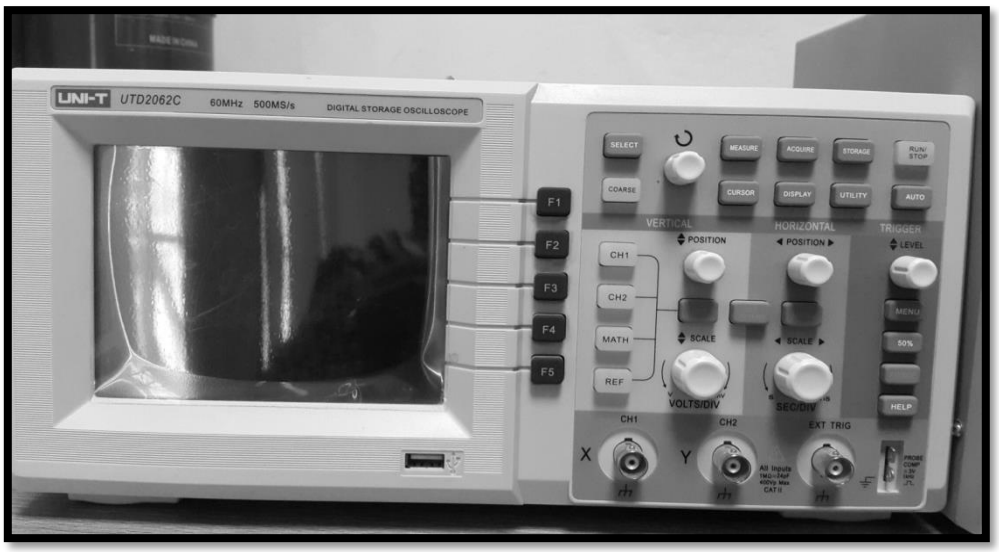
Digital Oscilloscope

Safety Precautions:

1- Avoid rough handling that leads to damaging the lab equipment.
2- Do not place any heavy object on oscilloscope

INTRODUCTION:

An oscilloscope is a laboratory instrument commonly used to display and analyze the waveform of electronic signals. In effect, the device draws a graph of the instantaneous signal voltage as a function of time. A typical oscilloscope can display alternating current (AC) or pulsating direct current (DC) waveforms having a frequency as low as approximately 1 hertz (Hz) or as high as several megahertz (MHz). High-end oscilloscopes can display signals having frequencies up to several hundred gigahertzes (GHz). The display is broken up into so-called horizontal divisions (hor div) and vertical divisions (vert div). Time is displayed from left to right on the horizontal scale. Instantaneous voltage appears on the vertical scale, with positive values going upward and negative values going downward.



What Can Scopes Measure:

In addition to those fundamental features, many scopes have measurement tools, which help to quickly quantify frequency, amplitude, and other waveform characteristics. In general a scope can measure both time-based and voltage-based characteristics:

- **Timing characteristics**:
  - **Frequency and period** – Frequency is defined as the number of times per second a waveform repeats. And the period is the reciprocal of that (number of seconds each repeating waveform takes). The maximum frequency a scope can measure varies, but it's often in the 100's of MHz (1E6 Hz) range.
  - **Duty cycle** – The percentage of a period that a wave is either positive or negative (there are both positive and negative duty cycles). The duty cycle is a ratio that tells you how long a signal is "on" versus how long it's "off" each period.
  - **Rise and fall time** – Signals can't instantaneously go from 0V to 5V, they have to smoothly rise. The duration of a wave going from a low point to a high point is called the rise time, and fall time measures the opposite. These characteristics are important when considering how fast a circuit can respond to signals.
- **Voltage characteristics**:

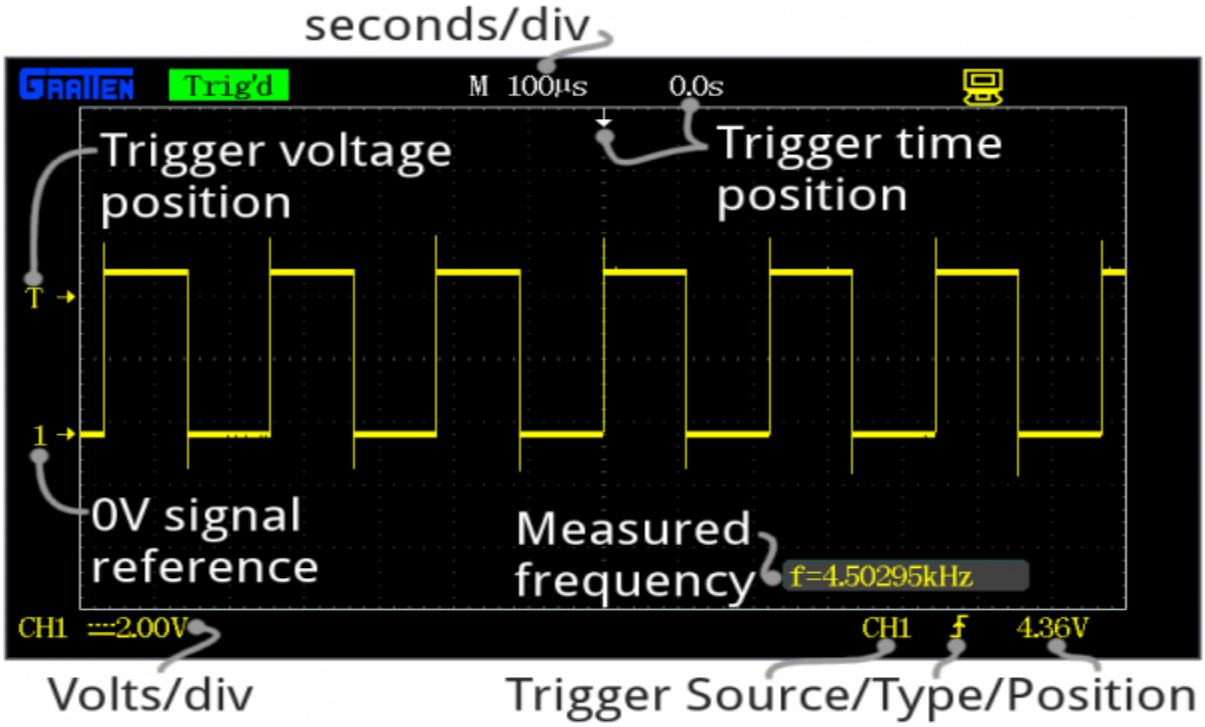
Amplitude – Amplitude is a measure of the magnitude of a signal. There are a variety of amplitude measurements including peak-to-peak amplitude, which measures the absoluted difference between a high and low voltage point of a signal. Peak amplitude, on the other hand, only measures how high or low a signal is past 0V.

  - **Maximum and minimum voltages** – The scope can tell you exactly how high and low the voltage of your signal gets.
  - **Mean and average voltages** – Oscilloscopes can calculate the average or mean of your signal, and it can also tell you the average of your signal's minimum and maximum voltage.

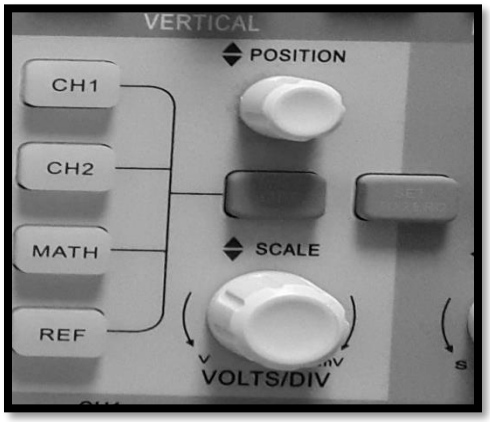
Front Control Panel of Oscilloscope

The Display

An oscilloscope isn't any good unless it can display the information you're trying to test, which makes the display one of the more important sections on the scope.



Every oscilloscope display should be criss-crossed with horizontal and vertical lines called divisions**.** The scale of those divisions is modified with the horizontal and vertical systems. The vertical system is measured in "volts per division" and the horizontal is "seconds per division". Generally, scopes will feature around 8-10 vertical (voltage) divisions, and 10-14 horizontal (seconds) divisions. Older scopes (especially those of the analog variety) usually feature a simple, monochrome display, though the intensity of the wave may vary. More modern scopes feature multicolor LCD screens, which are a great help in showing more than one waveform at a time. Many scope displays are situated next to a set of about five buttons – either to the side or below the display. These buttons can be used to navigate menus and control settings of the scope.



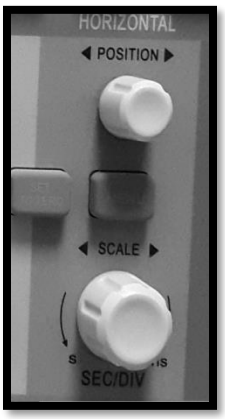
Vertical System

The vertical section of the scope controls the voltage scale on the display. There are traditionally two knobs in this section, which allow you to individually control the vertical position and volts/div. The more critical volts per division knob allows you to set the vertical scale on the screen. Rotating the knob clockwise will decrease the scale, and counter-clockwise will increase. A smaller scale – fewer volts per division on the screen – means you're more "zoomed in" to the waveform.

The position knob controls the vertical offset of the waveform on the screen. Rotate the knob clockwise, and the wave will move down, counter-clockwise will move it up the display. You can

use the position knob to offset part of a waveform off the screen. Using both the position and volts/div knobs in conjunction, you can zoom in on just a tiny part of the waveform that you care about the most. If you had a 5V square wave, but only cared about how much it was ringing on the edges, you could zoom in on the rising edge using both knobs.

Horizontal System



The horizontal section of the scope controls the time scale on the screen. Like the vertical system, the horizontal control gives you two knobs. Position and seconds/div. The seconds per division (s/div) knob rotates to increase or decrease the horizontal scale. If you rotate the s/div knob clockwise, the number of seconds each division represents will decrease – you'll be "zooming in" on the time scale. Rotate counter-clockwise to increase the time scale, and show a longer amount of time on the screen.

The position knob can move your waveform to the right or left of the display, adjusting the horizontal offset. Using the horizontal system, you can adjust how many periods of a waveform you want to see. You can zoom out, and show multiple peaks and troughs of a signal. Or you can zoom way in, and use the position knob to show just a tiny part of a wave.

Trigger System:-



The trigger section is devoted to **s**tabilizing and focusing the oscilloscope. The trigger tells the scope what parts of the signal to "trigger" on and start measuring. If your waveform is periodic**,** the trigger can be manipulated to keep the display static and unflinching. A poorly triggered wave will produce seizure-inducing sweeping waves like this. The trigger section of a scope is usually comprised of a level knob and a set of buttons to select the source and type of the trigger. The level knob can be twisted to set a trigger to a specific voltage point.

A series of buttons and screen menus make up the rest of the trigger system. Their main purpose is to select the trigger source and mode. There are a variety of trigger types, which manipulate how the trigger is activated:

- An edge trigger is the most basic form of the trigger. It will key the oscilloscope to start measuring when the signal voltage passes a certain level. An edge trigger can be set to catch on a rising or falling edge (or both).
- A pulse trigger tells the scope to key in on a specified "pulse" of voltage. You can specify the duration and direction of the pulse. For example, it can be a tiny blip of 0V -> 5V -> 0V, or it can be a seconds-long dip from 5V to 0V, back to 5V.
- A slope trigger can be set to trigger the scope on a positive or negative slope over a specified amount of time.

- More complicated triggers exist to focus on standardized waveforms that carry video data, like NTSC or PAL. These waves use a unique synchronizing pattern at the beginning of every frame.

You can also usually select a **triggering mode**, which, in effect, tells the scope how strongly you feel about your trigger. In automatic trigger mode, the scope can attempt to draw your waveform even if it doesn't trigger. **Normal mode** will only draw your wave if it sees the specified trigger. And **single mode** looks for your specified trigger, when it sees it it will draw your wave then stop.

For other auxiliary controls of oscilloscope's front panel students are instructed to read instruction manual of the oscilloscope.

Function Generator:-

A function generator is usually a piece of electronic test equipment or software used to generate different types of electrical waveforms over a wide range of frequencies. Some of the most common waveforms produced by the function generator are the sine, square, triangular and saw tooth shapes but here in physics lab we will use audio generator



Write down the function of following controls of Audio generator as explained during demonstration

1- Power Button:

2- Power Out:

3- Amplitude Knob:

4- Waveform:

5- Freq Range Selection:

6- Freq knob:

7- BNC Output:

8- Attenuators(db):

9- BNC Input:

10- Ext button

11- Dsiplay:

Task: Signal generation and display on Oscilloscope

Generate signals of varying amplitude and frequencies using signal generator and, measure different parameters using the given scales, cursors andusing the automated measurement function.

	frequency	Vp-p	Graticule		Cursors		Measurement Tool	
			Vrms	T	Vrms	T	Vrms	T
1	500HZ	2V						
2	2KHz	3V						
3	5KHz	5V						

Conclusion and Remarks:

Results :



College of Electrical & Mechanical

Engineering, NUST



Department of Basic Sciences & Humanities

Applied Physics

Lab No 10

Determination of RC time constant of RC circuit

Name……..

Degree & Discipline……..

Group No…….

Date……

Date of Submission……

WORKING (60%)	PROBLEM UNDERSTANDING (20%)	RESULTS (20%)

Determination of RC time constant of RC circuit

Objective:

1. To study charging and discharging of capacitor in RC circuit.
2. To determine charging time constant

Equipment Required:-

1. Audio Generator
2. Digital Oscilloscope
3. Components R = 1KΩ, C =1μF

Safety Precautions:

1- Avoid rough handling that leads to damaging the lab equipment.
2- Do not place any heavy object on oscilloscope

INTRODUCTION:

Theory: Charging RC Circuits

A capacitor stores separated charges for later use. The capacitance C of a capacitor is defined as the absolute value of the ratio of the charge Q on either plate to the potential difference VC between the plates,

$$C = \frac{Q}{V} \qquad (1)$$

with units of C/V ≡ F (farads). A farad is a very large unit; typical capacitors used in electronic circuits have values in the range of μF, nF, or even pF. The process of charging up a capacitor from a power supply takes time. This time depends on the overall capacitance C and resistance R in a circuit. Consider, for example, the series RC circuit shown in Fig. 1.

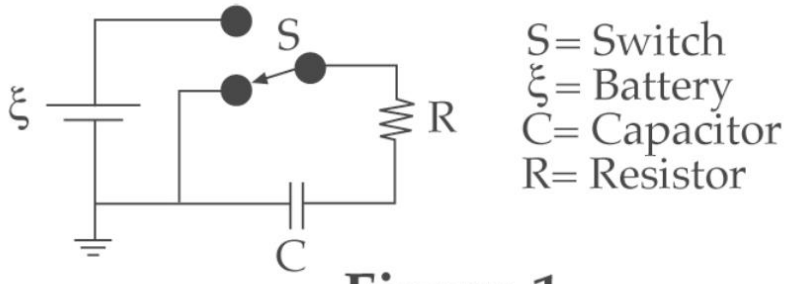


Figure 1

If the switch is turned on at time t = 0, the charge Q on the positive plate of the capacitor will increase from zero initially to some final value which we'll call Q∞. To determine this final value and to find how rapidly the charge approaches it, we can apply Kirchhoff's Voltage (Loop) Rule to the circuit of Fig. 1. If VR is the voltage drop across the resistor and VC is the voltage drop across the capacitor, then the sum of these two drops must equal the voltage rise ξ across the battery, so that

$$\xi = V_C + V_R \qquad (2)$$

However, by Ohm's law VR = IR where the current in the circuit equals the rate of flow of charge, I = ΔQ/Δt. Making these substitutions in Eq. (2) along with the value of VC from Eq. (1) gives

$$\xi = R\frac{\Delta Q}{\Delta t} + \frac{Q}{C} \qquad (3)$$

$$Q = Q_\infty\left(1 - e^{-t/RC}\right) \qquad (4)$$

Where $Q_\infty = C\xi$ is the final charge on the positive plate of the capacitor? This value of Q∞ makes intuitive sense: after the circuit has settled down and current has stopped flowing, there is no voltage drop across the resistor (because of Ohm's law) and hence the entire battery voltage ξ appears across the capacitor, so that $Q_\infty = C\xi$ from Eq. (1). Note in Eq. (4) that τ ≡ RC has SI units of seconds (if R is in Ω and C is in F) and is called the time constant of the circuit.

Notice from Eq. (4) that the capacitor attains 63.2% of its final charge in one time constant, because

$$t = \tau \Rightarrow \frac{Q}{Q_\infty} = 1 - e^{-1} = 0.632 \qquad (5)$$

We can use Eq. (6) to find the voltages and current in the circuit. Substituting Eq. (4) into (1) implies that

$$V_C = \xi\left(1 - e^{-t/\tau}\right) \qquad (6)$$

Using KVL, we get

$$V_R = \xi e^{-t/\tau} \qquad (7)$$

Task1: Measure the RC time constant of the RC circuit using the charging and discharging of capacitor.

- Construct RC circuit on the breadboard
- Calculate RC time constant using the given component values
- Set the frequency 10 times the time constant value.
- Set the square wave signal waveform of 8V Vp-p.
- Connect channel CH-1 of the oscilloscope across source and CH-2 across capacitor to display Vin and Vout=V_C.
- Measure time t= $T_{1/2}$ at which VC=0.5Vo
- Calculate RC using the expression $V_c = V_o \exp(-\ t/RC)$
- Draw signal on graph.

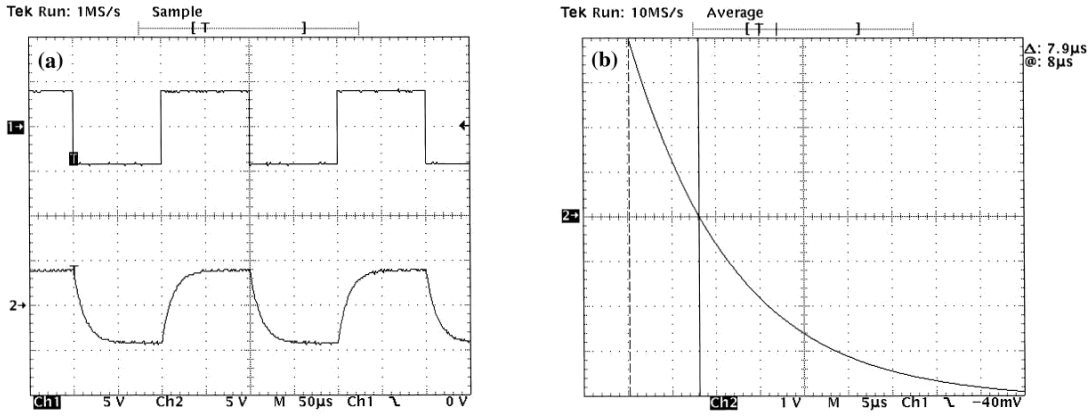
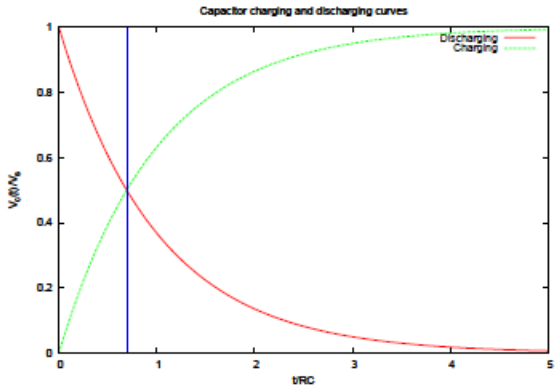


Figure 1 Figure Digital scope measurements of RC decay. (a) Display showing both the input square wave (CH1) and the output waveform that is rounded due to RC charging (CH2). (b) Expanded view of the signal in CH2 showing the cursors placed to measure one half-life time of the decay.



- Figure 2: The capacitor charging and discharging curves. The vertical blue line is the "half life" point of the charging and discharging timeline.

Task2: Measure the RC time constant of the RC circuit using AC response of the capacitor.

- Set the signal generator to output sinusoidal signal waveform of 10V Vp-p.
- Connect channel CH-1 of the oscilloscope across source and CH-2 across capacitor to display Vin and Vout=V_C.
- Set the frequency 0Hz and start increasing the frequency and observe the changes in capacitor voltage VC with frequency. Record your observation to discuss in your lab report.
- Record frequency at which VC=0.707Vo.
- Calculate RC using the expression
- At critical frequency

  $R = X_C$, where $X_C = \frac{1}{2\pi fC}$ hence $RC = \frac{1}{2\pi f_C}$, where f_C is critical frequency

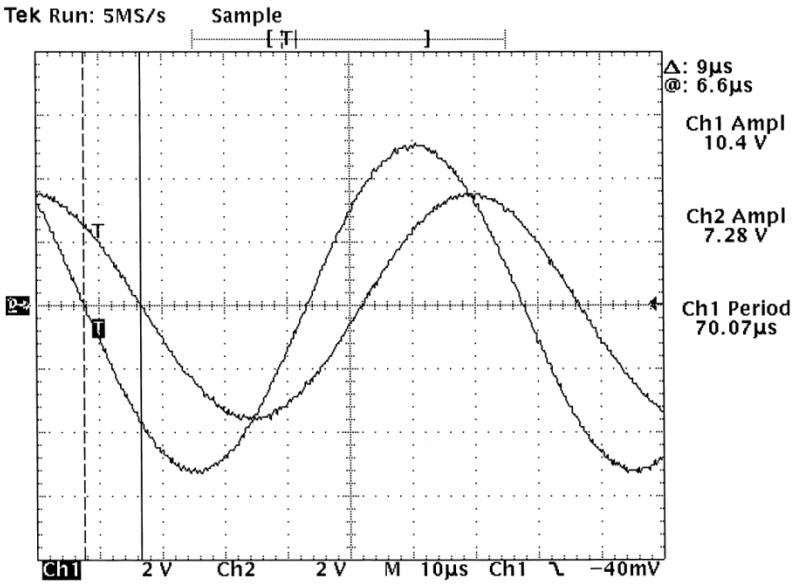


Figure 2 Figure 6: Digital scope measurements of sine wave response of an RC circuit. The larger waveform is the signal at the input of the RC circuit, and the smaller waveform is the signal at the output. The frequency of the sine wave is set so that the ratio of the output to input amplitudes is about 0.7.

Results :

Questions

1. ?

   Ans……
   ……
   ……
   ……
   ……..

2. ?

   Ans……
   ……
   ……
   ……
   ……..

3. ??

   Ans……
   ……
   ……
   ……
   ……..

4. ?

   Ans……
   ……
   ……
   ……
   ……..

5. ?

   Ans……
   ……
   ……
   ……
   ……..

College of Electrical & Mechanical Engineering, NUST

Department of Basic Sciences & Humanities

Applied Physics

Lab No 11

Investigation of frequency response (VC, XC) of capacitor in RC circuit

Name……..

Degree & Discipline……..

Group No…….

Date……

Date of Submission……

WORKING (60%)	PROBLEM UNDERSTANDING (20%)	RESULTS (20%)

Investigation of frequency response (VC, XC) of capacitor in RC circuit

Objective:

1. To study capacitive reactance as function of frequency.
2. To study capacitive voltage in RC circuit as function of frequency.

Equipment

1. AC Function Generator model:_____srn:____
2. DMM model:_____srn:____

Components

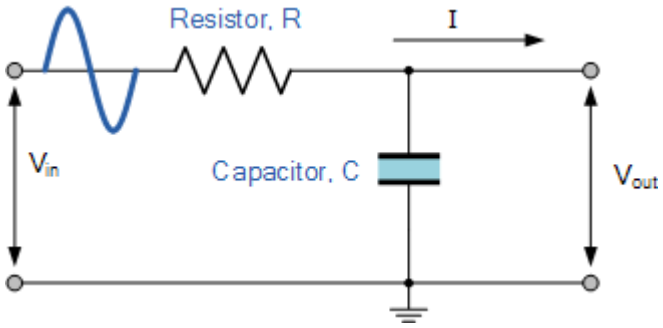
a. 1µF actual:__
b. 100 Ohm actual:__

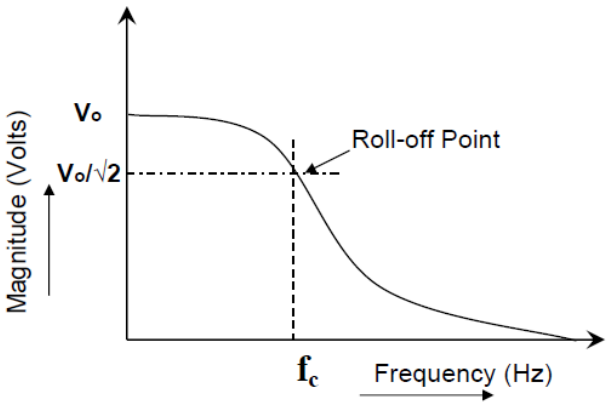
Introduction:

The current – voltage characteristic of a capacitor is unlike that of typical resistors. While resistors show a constant resistance value over a wide range of frequencies, the equivalent ohmic value for a capacitor, known as capacitive reactance, is inversely proportional to frequency. The capacitive reactance may be computed via the formula:

$$XC = \frac{1}{2\pi f C}$$

Low Pass Filter:





A simple passive RC Low Pass Filter or LPF, can be easily made by connecting together in series a single Resistor with a single Capacitor as shown below. In this type of filter arrangement the input signal (Vin) is applied to the series combination (both the Resistor and Capacitor together) but the output signal (Vout) is taken across the capacitor only.

The capacitive reactance has inverse relation with frequency, as frequency increases, capacitive reactance decreases. At low frequencies, the capacitive reactance dominates over the resistance. Therefore, the signal voltage is dropped off mostly across the capacitor. It is as though the capacitor is offering more effective resistance than the resistor. At high frequencies, the capacitive reactance becomes negligible; most of the signal voltage is across the resistor. . In the extreme case, at infinite frequency, the reactance is zero; the current is maximum, all the voltage is across the resistor. At high frequency, the capacitive reactance is zero, and it is as though there is no capacitor in the circuit, or the capacitor is shorted out. These statements are generally summed up by saying that the capacitor acts like a block for low frequencies, but like a short for high frequencies. At corner or critical frequency, capacitive reactance becomes equal to series resistance in the circuit hence we can calculate critical frequency of the circuit using relation

$$X_C = R$$

$$\frac{1}{2\pi f_c C} = R$$

$$f_C = \frac{1}{2\pi RC}$$

Procedure

1. Set the generator for the sine wave and frequency of 1000Hz.
2. Measure the values of given components using DMM and record the data.
3. Measure the output of audio generator using DMM and set it to 2V RMS.
4. Connect the circuit as shown in the figure.
5. Calculate the theoretical value of Xc using the measured capacitor value for each frequency and record in Table 11.1.
6. Record the RMS capacitor voltage and resistor voltage for each frequency and record in Table 11.1.
7. Determine the source current using ohms law for resistor voltage and record in table 11.1 .
8. Using source current and capacitor voltage determine the capacitive reactance and record in the table11.1.
9. Plot graph between AC reactance of capacitor against frequency.
10. Plot Vout vs frequency for the RC circuit and verify cut in frequency
11. Plot Xc vs frequency for the RC circuit and verify cut in frequency.

Observation and Calculation

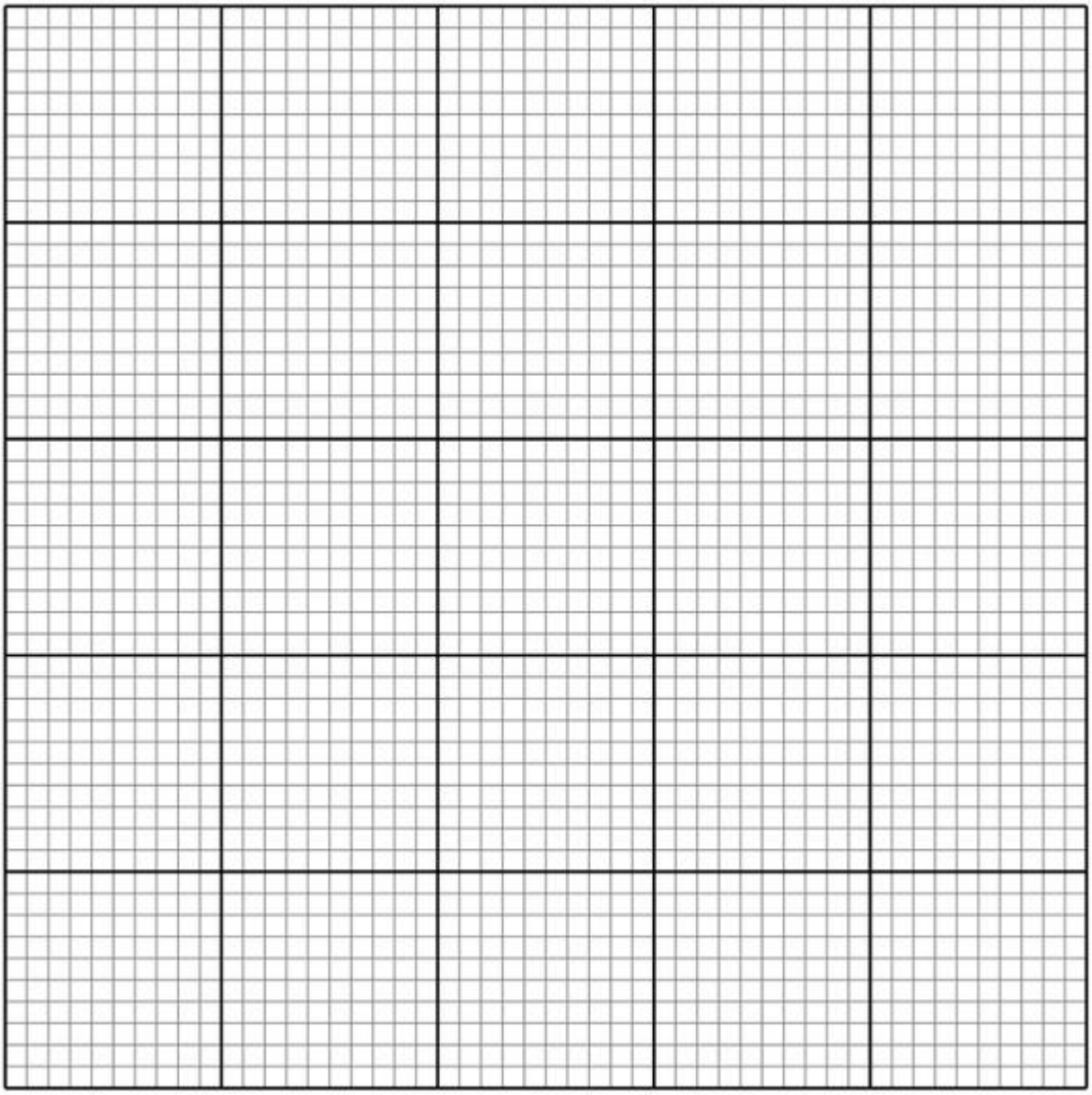
Measurements:

Observed Data:-

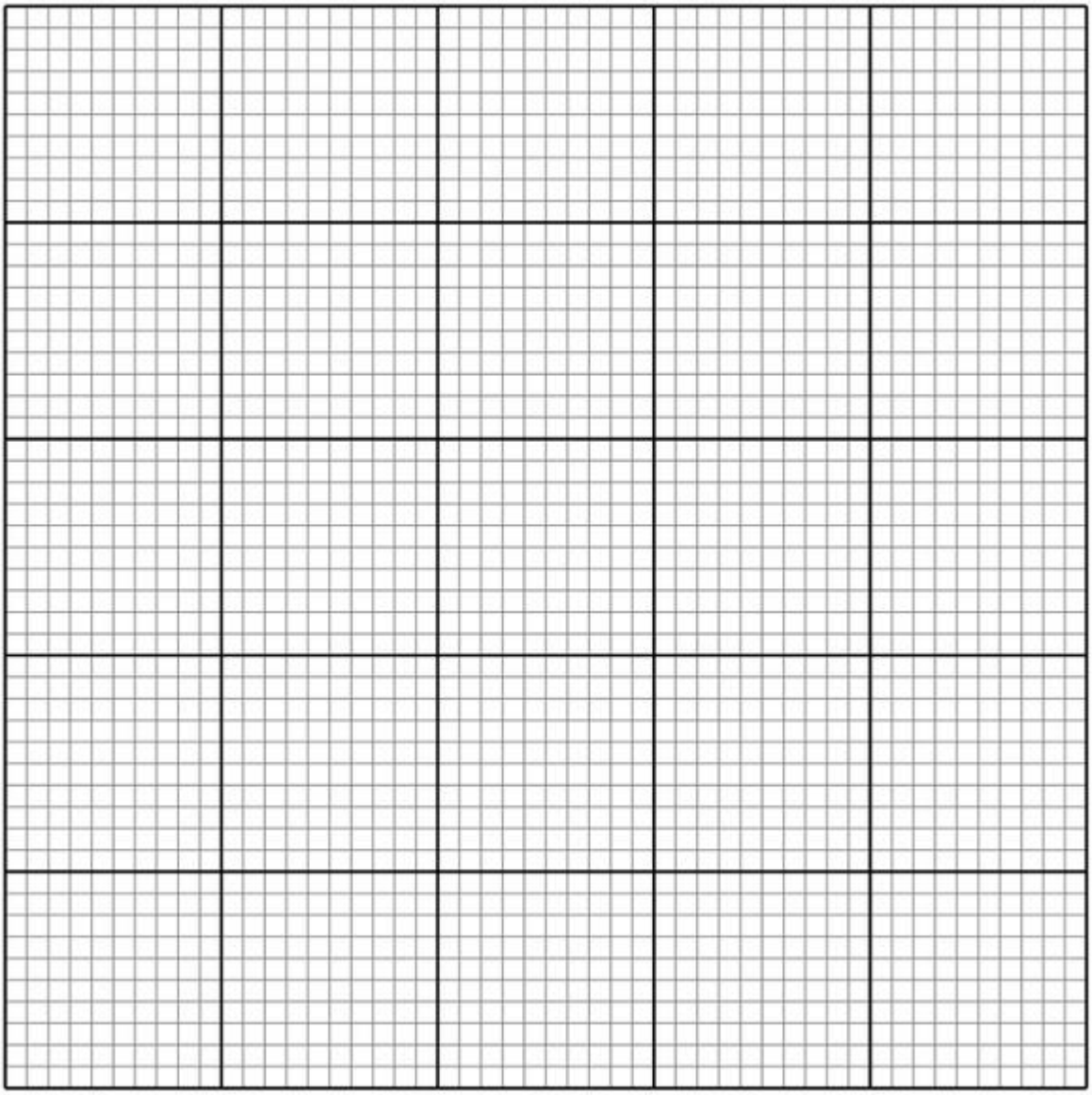
Table 11.1

Frequency	V_R	V_C	$I=V_R/R$	$X_C=V_C/I$

APPLIED PHYSICA PHY-102



APPLIED PHYSICA PHY-102



Questions

1. What is the relationship between capacitive reactance and frequency?

   Ans……
   ……
   ……
   ……
   ……..

2. What is the behavior of capacitor at low and high frequencies?

   Ans……
   ……
   ……
   ……
   ……..

3. If the experiment had been repeated with frequencies 10 times higher than those used in Table 10.1, what would the resulting plots look like?

   Ans……
   ……
   ……
   ……
   ……..

4. If the experiment had been repeated with frequencies 10 times lower than those in Table 10.1, what effect would that have on the experiment?

   Ans……
   ……
   ……
   ……
   ……..

5. Define critical frequency.

   Ans……
   ……
   ……
   ……
   ……..

Results :

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Engineering, NUST

Department of Basic Sciences & Humanities

Applied Physics

Lab No 12

Determination of Thermal Coefficient of Linear expansion for different metals.

Name……..

Degree & Discipline……..

Group No…….

Date……

Date of Submission……

WORKING (60%)	PROBLEM UNDERSTANDING (20%)	RESULTS (20%)

Determination of Thermal Coefficient of Linear expansion for different metals.

Objective: To Measure the Coefficient of Linear Expansion for Aluminum, Brass, and Copper.

Apparatus :

- Thermal Expansion Apparatus
- Steam Generator
- Container (to catch water)
- Meter Stick or Measuring Tape



Theory :

Most materials expand somewhat when heated through a temperature range that does not produce a change in phase. The added heat increases the average amplitude of vibration of the atoms in the material, which increases the average separation between the atoms. Suppose an object of length L undergoes a temperature change of magnitude ΔT. If ΔT is reasonably small, the change in length, ΔL, is generally proportional to L and ΔT.

Stated mathematically: $\Delta L = \alpha L \Delta T$

where α is called the coefficient of linear expansion for the material. For materials that are not isotropic, such as an asymmetric crystal for example, α can have a different value depending on the axis along which the expansion is measured. The coefficient α can also vary somewhat with temperature so that the degree of expansion depends not only on the magnitude of the temperature

change, but on the absolute temperature as well. In this experiment, you will measure α for aluminum, brass, and copper. These metals are isotropic so that a need only be measured along one dimension. Also, within the limits of this experiment, a does not vary with temperature.

Notes on Temperature Measurement

A thermistor's resistance varies reliable with temperature. The resistance can be measured with an ohmmeter and converted to a temperature measurement using the conversion table provided on the tag attached to the apparatus and also found at the end of this manual. Although the relationship between temperature and resistance is not linear, a linear approximation can be accurately used to interpolate between table data points with an accuracy of approximately ±0.2°C. The thermistor used to measure the tube temperature is embedded in that thermistor lug that is fastened to the tube. Once thermal equilibrium has been reached, the heat is highly uniform along the length of the tube. The foam insulation is used to inhibit heat loss through the thermistor lug so the lug temperature closely follows the tube temperature. The insulator does not have an appreciable effect on the local temperature of the tube itself.

Procedure:

1. Measure L, the length of the aluminum tube at room temperature. Measure from the inner edge of the larger circular disk on one end to the inner edge of the smaller circular disk at the other end (see Figure 1). Record your results in Table 1.

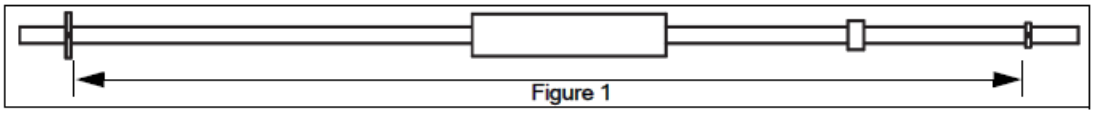


Figure 1

2. Mount the aluminum tube in the apparatus frame as shown in Figure 2. The smaller circular disk on the tube fits into a slot on the "high" end frame. The larger circular disk on the tube presses against the tip of the spring arm of the dial indicator.

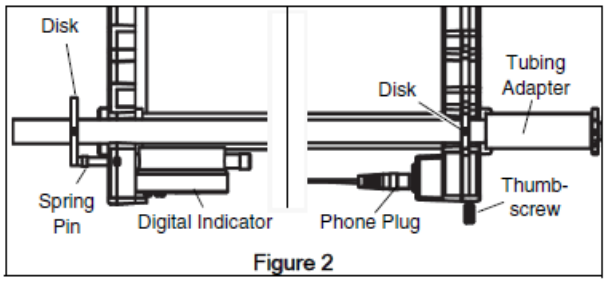


Figure 2

3. Turn the metal tuabe so that the thermistor lug under the foam insulation is on top. Connect the phone plug on the end of the thermistor cable into the phone port on the "high" end frame, or into the phone jack on a compatible PASDCO sensor.
4. Tighten the thumbscrew in the "high" end frame against the tube until it can no longer be

moved.

5. Make sure that the foam insulator is centered over the thermistor lug.
6. If you are used an ohmmeter (or multimeter), plug the leads of your ohmmeter into the banana plug connectors on the "high" end frame just under the phone plug port.
7. Measure and record Rrm, the resistance of the thermistor at room temperature. Record this value in the table.
8. Attach the tubing adapter of the rubber tubing to the end of the aluminum tube. (Attach it to the end farthest from the digital indicator.) Connect the other end of the tubing to the steam generator.
9. Place a container under the other end of the tube to catch the draining water that condenses in the tube.
10. Press the ON/OFF button on the digital indicator to turn it on. Press the ZERO button to set the initial digital reading to zero. As the tube expands, the spring pin of the digital indicator will stay in contact with the larger circular disk on the tube.
11. Turn on the steam generator. As steam begins to flow, watch the digital display and the resistance reading on the ohmmeter (or computing device). When the thermistor resistance stabilizes, record the resistance (Rhot) in Table 1. Also, record the expansion of the tube length (ΔL) as indicated by the display on the digital indicator.
12. Repeat the experiment for the other two metal tubes.

Data and Calculations

Table 1: Results

	Data				Calculations		
Tube	L (mm)	Rrm (Ω)	ΔL (mm)	Rhot (Ω)	Trm (°C)	Thot (°C)	ΔT (°C)
Aluminum							
Brass							
Copper							

1. Use the Conversion Table the one attached to the frame tube of the apparatus, to convert your thermistor resistance measurements, Rrm and Rhot, into temperature measurements, Trm and Thot. Record your results in the table.
2. Calculate ΔT = Thot - Trm. Record the result in the table.
3. Using the equation

$$\Delta L = \alpha L \Delta T,$$

calculate α for aluminum, brass, and copper.

- Aluminum = _____
- Brass = _____
- Copper = _____

Accepted Values for the Coefficient of Thermal Expansion,

Material	α_L (x 10^{-6}/°C)	Composition (%)
Aluminum (6061-T6)	23.6	Al (95.8 - 98.6), Mg (0.80 - 1.2), Fe (<0.70), Si (0.40 - 0.80)
Brass (C270)	20.3	Brass (63 - 68.5), Zn (31.3 - 37), Pb (0.1), Fe (0.07)
Copper (C122)	17.0	Cu (99.9), P (0.02 nominal)

Questions

1. From your result, can you calculate the coefficient of volume expansion for aluminum, brass, and copper?

$$\Delta V = \alpha_{vol} V \Delta T$$

Ans……
……
……
……
……..

College of Electrical & Mechanical

Engineering, NUST

Department of Basic Sciences & Humanities

Applied Physics

Lab No 13

To study the PN junction diode characteristics under Forward & Reverse bias conditions.

Name……..

Degree & Discipline……..

Group No…….

Date……

Date of Submission……

WORKING (60%)	PROBLEM UNDERSTANDING (20%)	RESULTS (20%)

To study the PN junction diode characteristics under Forward & Reverse bias conditions.

Objective: To study the PN junction diode characteristics under Forward & Reverse bias conditions.

Apparatus :

- Power Supply (0-30) V 1
- Ammeter (0–100) μA 1
- Voltmeter (0–1) V 1
- Diode IN4001 1
- Resistor 1kΩ 1
- connecting wires.

Theory :

A PN junction diode is a two terminal junction device. It conducts only in one direction (only on forward biasing).

FORWARD BIAS:

On forward biasing, initially no current flows due to barrier potential. As the applied potential exceeds the barrier potential the charge carriers gain sufficient energy to cross the potential barrier and hence enter the other region. The holes, which are majority carriers in the P-region, become minority carriers on entering the N-regions, and electrons, which are the majority carriers in the N-region, become minority carriers on entering the P-region. This injection of Minority carriers results in the current flow, opposite to the direction of electron movement.

REVERSE BIAS:

On reverse biasing, the majority charge carriers are attracted towards the terminals due to the applied potential resulting in the widening of the depletion region. Since the charge carriers are pushed towards the terminals no current flows in the device due to majority charge carriers. There will be some current in the device due to the thermally generated minority carriers. The generation of such carriers is independent of the applied potential and hence the current is constant for all increasing reverse potential. This current is referred to as Reverse Saturation Current (Io) and it increases with temperature. When the applied reverse voltage is increased beyond the certain limit, it results in breakdown. During breakdown, the diode current increases tremendously.

PROCEDURE: FORWARD BIAS:

1. Connect the circuit as per the diagram.
2. Vary the applied voltage V in steps of 0.1V.
3. Note down the corresponding Ammeter readings I_f in Table 13.1.
4. Plot a graph between V_f and I_f.
5.

OBSERVATIONS

1. Find the DC(static) resistance = Vf / If.=
2. Find the AC (dynamic) resistance $r = \frac{\partial V}{\partial I} = \frac{\Delta V}{\Delta I}$ _____
3. Find the forward voltage drop = [Hint: it is equal to 0.7 for Si and 0.3 for Ge]=

REVERSE BIAS:

1. Connect the circuit as per the diagram.
2. Vary the applied voltage Vr in steps of 0.5V.
3. Note down the corresponding Ammeter readings Ir in Table 13.1.
4. Plot a graph between Vr & Ir
5. Find the dynamic resistance $r = \frac{\partial V}{\partial I} = \frac{\Delta V}{\Delta I}$

Specification for 1N4001: Silicon Diode

Peak Inverse Voltage: 50V

Maximum forward voltage drop at 1 Amp is 1.1 volts

Maximum reverse current at 50 volts is 5µA

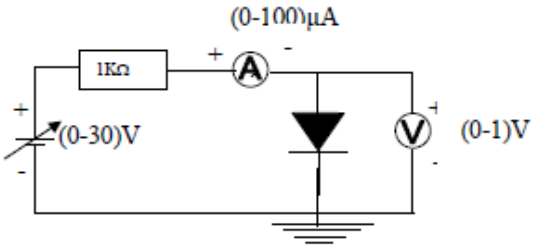
RESULT:

Forward and Reverse bias characteristics of the PN junction diode was Studied and the dynamic resistance under

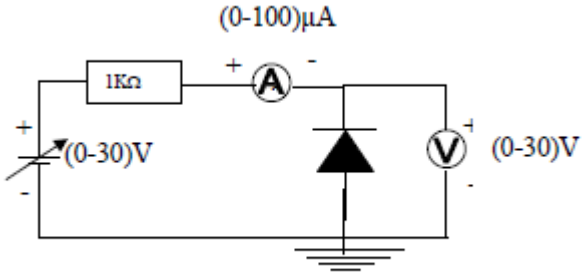
Forward bias = ----------, Reverse bias = ----------.

CIRCUIT DIAGRAM:

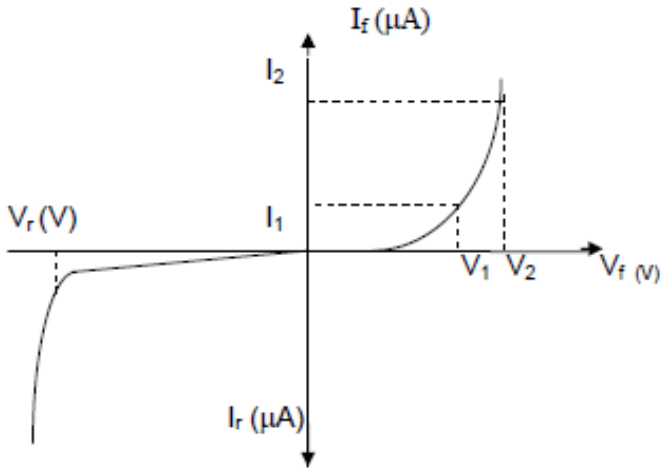
FORWARD BIAS:



REVERSE BIAS:



MODEL GRAPH



Observation and Calculation

Observed Data:-

FORWARD BIAS:

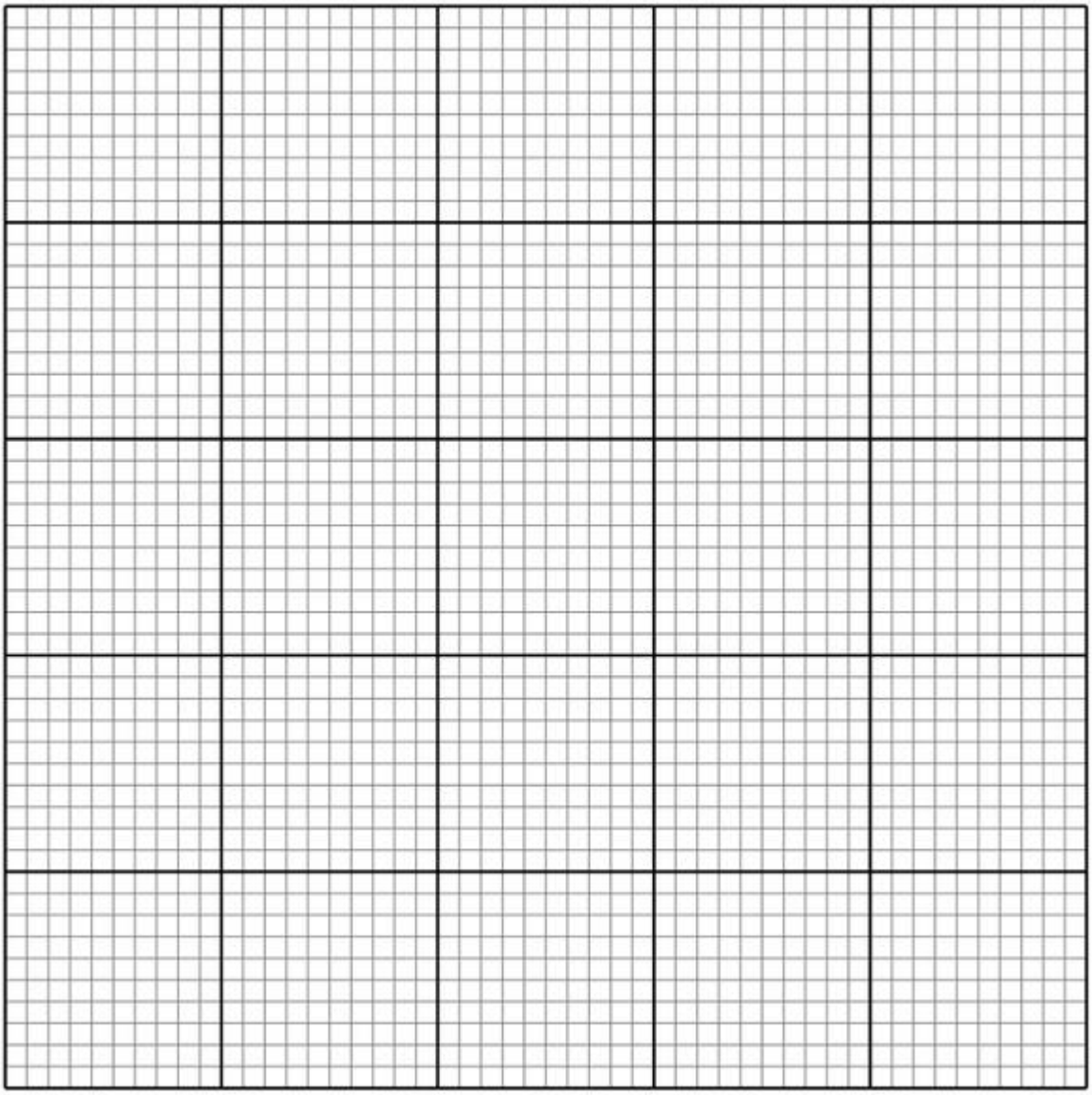
Table 13.1

S.No.	VOLTAGE (Vf) (In Volts)	CURRENT (If) (In μA)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		

REVERSE BIAS:

S.No.	VOLTAGE (Vr) (In Volts)	CURRENT (Ir) (In μA)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		

APPLIED PHYSICA PHY-102



Questions

1. What happens when we join ***p***-type and ***n***-type semiconductors?

   Ans……
   ……
   ……
   ……
   ……..

2. What is the width of the depletion region? How does it relate to the dopant concentration?

   Ans……
   ……
   ……
   ……
   ……..

3. What happens when we apply voltage to the ***p-n*** junction? What is forward and reverse bias?.

   Ans……
   ……
   ……
   ……
   ……..

4. What is the current-voltage characteristic for the ***p-n*** junction diode? Why is it different from a resistor?

   Ans……
   ……
   ……
   ……
   ……..

5. Define majority carriers and minority carriers, Define reverse leakage current.

   Ans……
   ……
   ……
   ……
   ……..

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Applied Physics

Lab No 14

Hall Effect: Study the Hall voltage relationship with magnetic field and current.

Name……..

Degree & Discipline……..

Group No…….

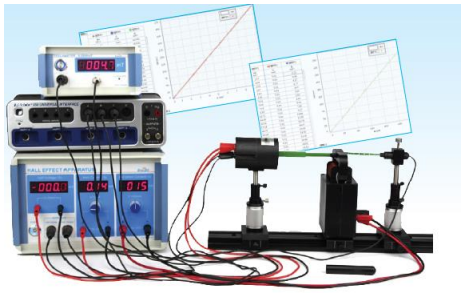
Date……

Date of Submission……

WORKING (60%)	PROBLEM UNDERSTANDING (20%)	RESULTS (20%)

Hall Effect: Study the Hall voltage relationship with magnetic field and current.

Objective: To study the Hall Effect



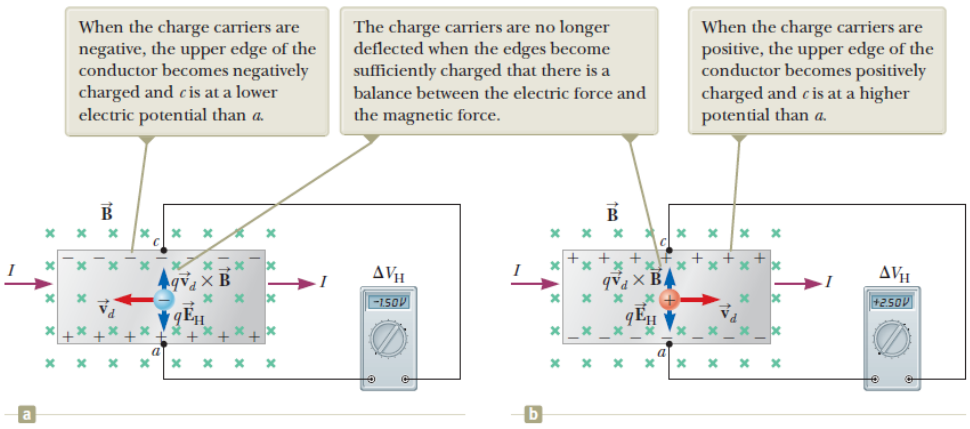
Apparatus :

- Hall Probe Unit, n-Semiconductor (GaAs)
- Hall Effect Apparatus
- Tesla meter

Theory :

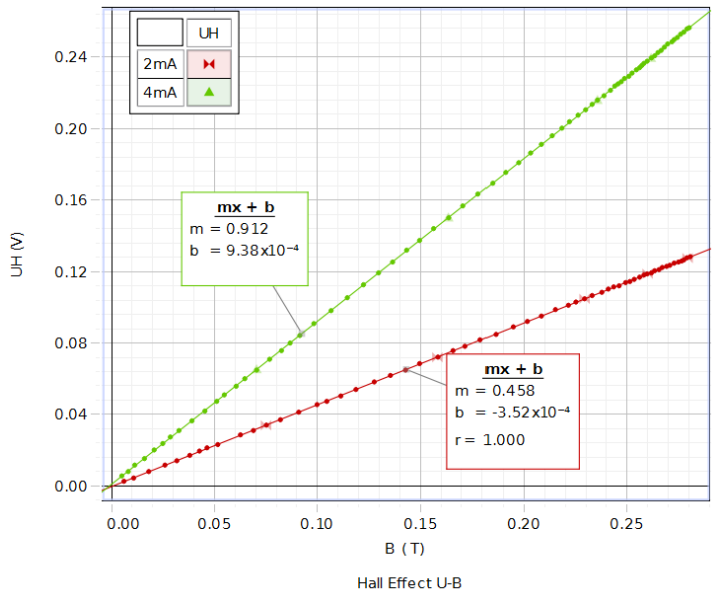
When a current-carrying conductor is placed in a magnetic field, a potential difference is generated in a direction perpendicular to both the current and the magnetic field. This phenomenon, first observed by Edwin Hall (1855–1938) in 1879, is known as the *Hall effect*. The arrangement for observing the Hall effect consists of a flat conductor carrying a current I in the x direction as shown in Figure. A uniform magnetic field **B** is applied in the y direction. If the charge carriers are electrons moving in the negative x direction with a drift velocity vd, they experience an upward magnetic force FB=q vX B, are deflected upward, and accumulate at the upper edge of the flat conductor, leaving an excess of positive charge at the lower edge Fig. This accumulation of charge at the edges establishes an electric field in the conductor and increases until the electric force on carriers remaining in the bulk of the conductor balances the magnetic force acting on the carriers. When this equilibrium condition is reached, the electrons are no longer deflected upward. A sensitive voltmeter connected across the sample as shown in Figure can measure the potential difference, known as the Hall voltage ΔV_H generated across the conductor.

$$\Delta V_{\mathrm{H}} = \frac{IBd}{nqA}$$



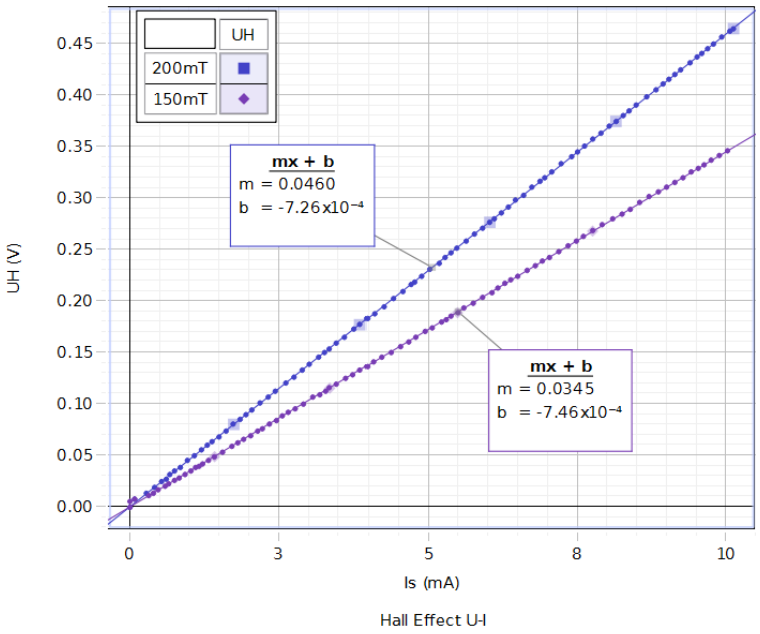
TASK A:

When the current is constant, research the relationship between Hall voltage and magnetic field.



TASK B:

When the magnetic field is constant, research the relationship between Hall voltage and Hall current.



Observation and Calculation

Table 14.1

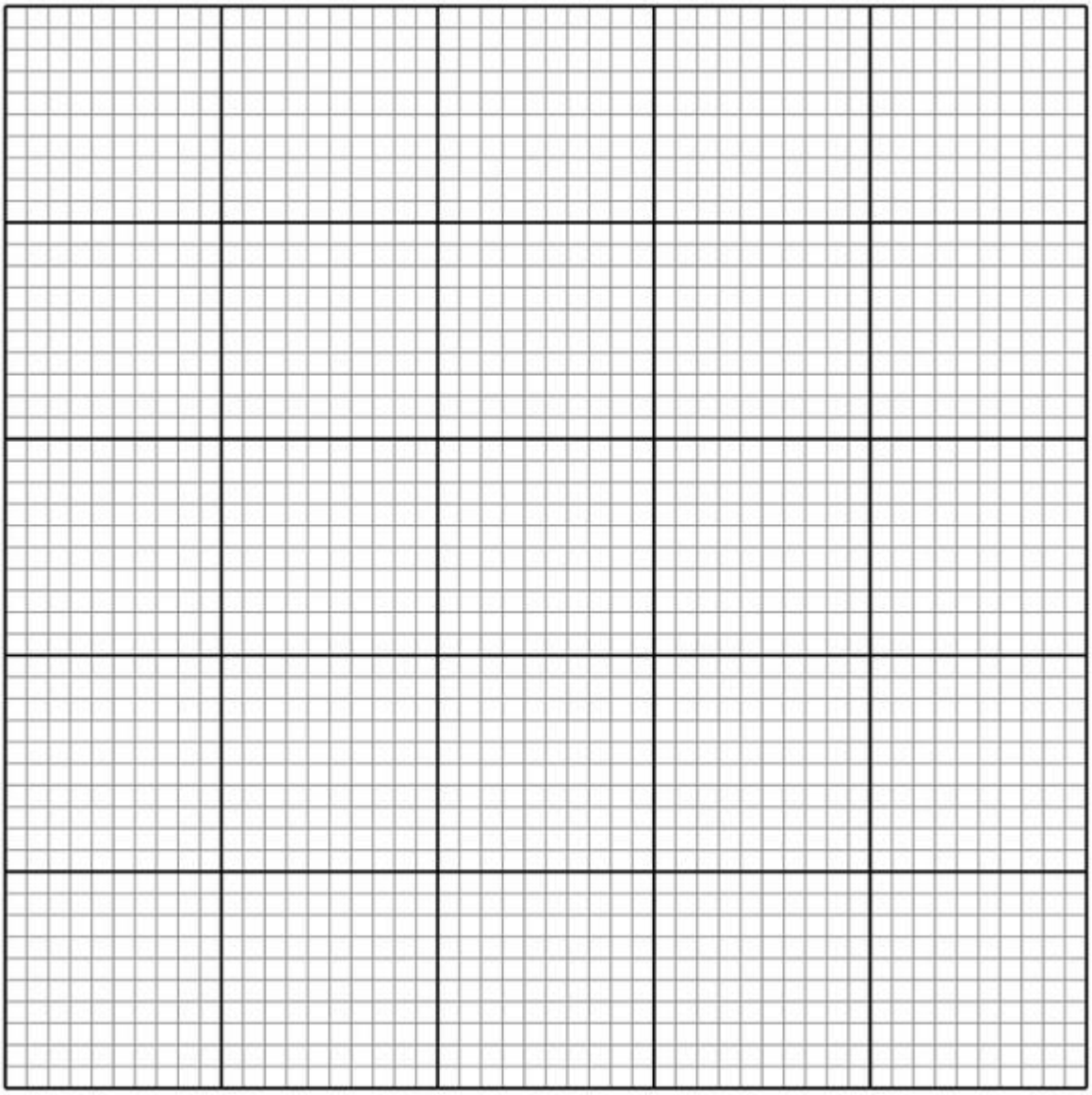
Task A

S.No.	Hall Voltage ΔV_H (Volts)	Magnetic Field B (Tesla)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		

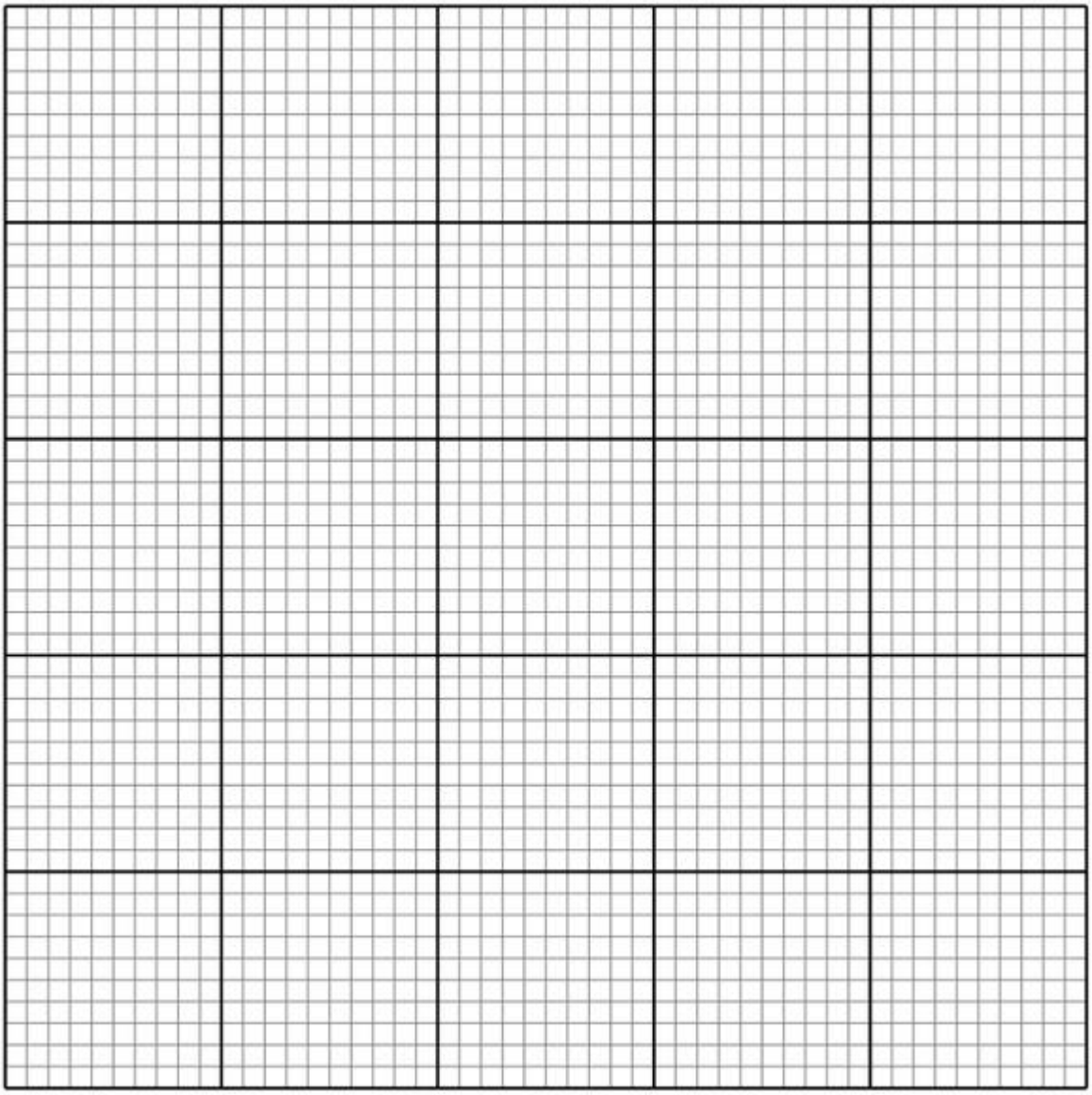
Task B

S.No.	Hall Voltage ΔV_H (Volts)	Hall Current (mA)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		

APPLIED PHYSICA PHY-102



APPLIED PHYSICA PHY-102



Questions

1. Define Hall Effect and Hall current.

   Ans……
   ……
   ……
   ……
   ……..

2. Derive expression for Hall Voltage.

   Ans……
   ……
   ……
   ……
   ……..

3. Derive expression for unknown charge density.

   Ans……
   ……
   ……
   ……
   ……..

4. What does the slope of graph a represents?

   Ans……
   ……
   ……
   ……
   ……..

5. What does the slope of graph b represents?

   Ans……
   ……
   ……
   ……
   ……..